



21st International Stirling Engine Conference (ISEC 2026)

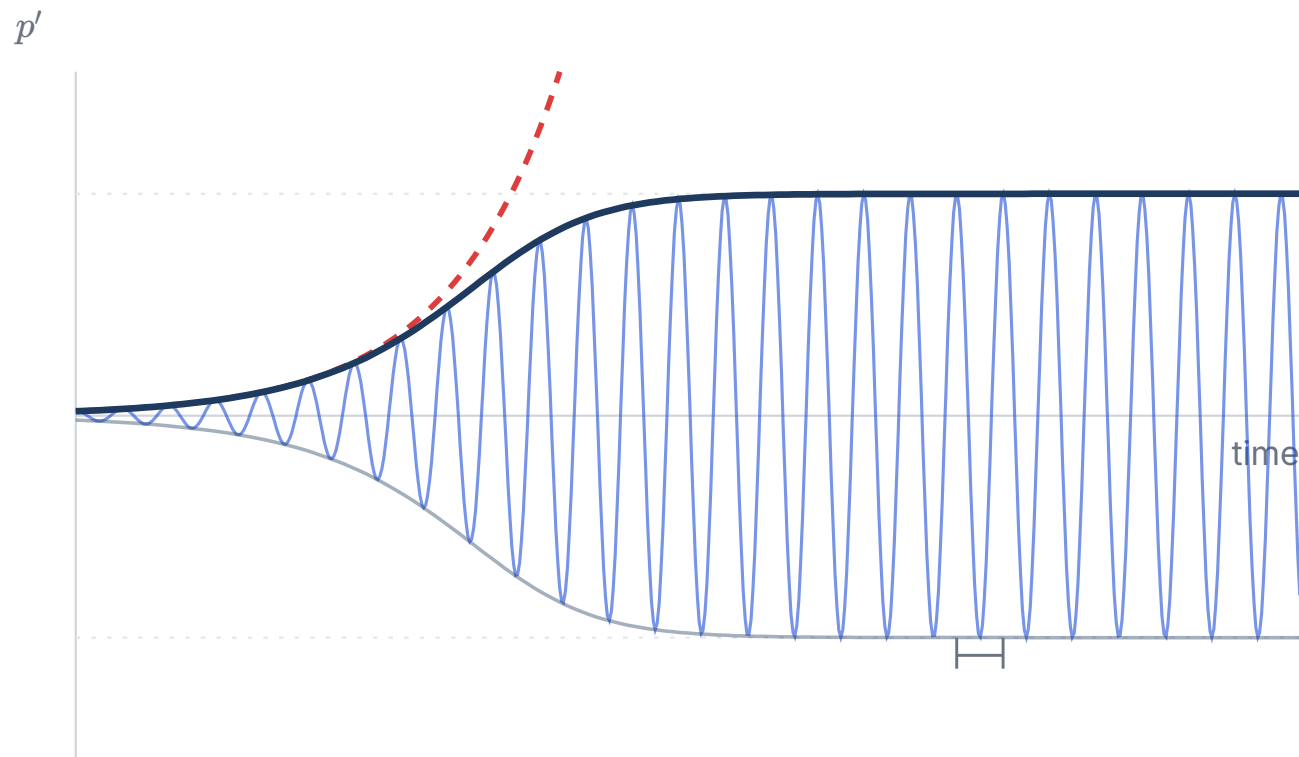
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Operator-based analysis of a loudspeaker-coupled thermoacoustic Stirling engine

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A working engine lives in saturation



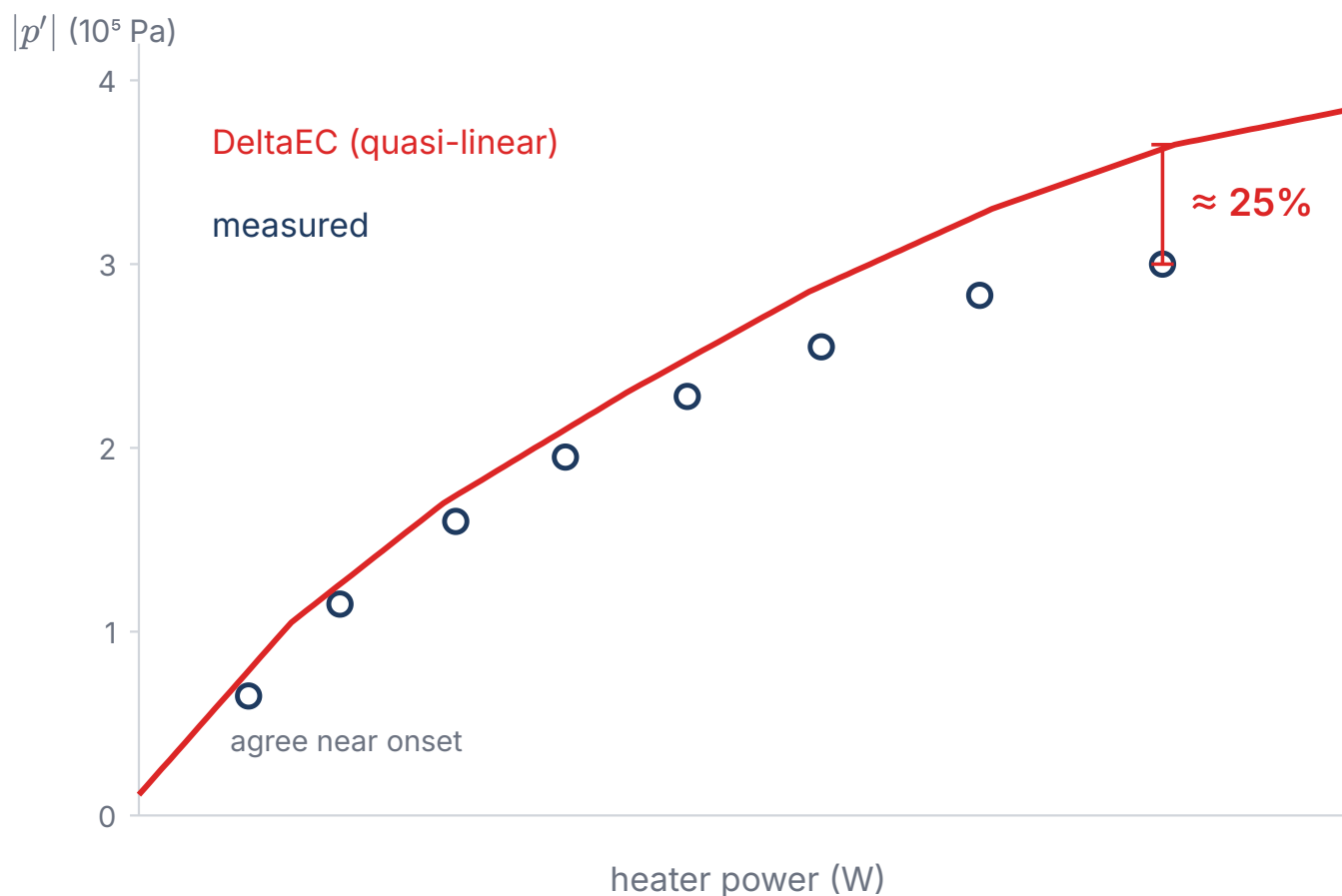
Envelope of the acoustic pressure after crossing the Hopf threshold (schematic).

$$p'(x, t) = \Re\{P(x) e^{i\sigma t}\}, \quad i\sigma = \sigma_r + i\omega$$

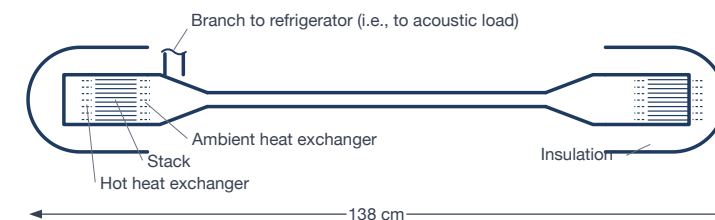
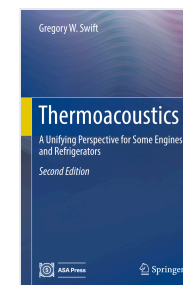
$$e^{i\sigma t} = \underbrace{e^{\sigma_r t}}_{\text{growth}} \underbrace{e^{i\omega t}}_{\text{oscillation}} \Rightarrow |p'| \propto e^{\sigma_r t}$$

- Linear theory hands you one complex **eigenvalue**: σ_r sets the growth, ω sets the frequency.
- With σ_r fixed and positive, it predicts the oscillation **grows forever**.
- In the real engine, the growth rate is amplitude-dependent and **falls to zero**, and the oscillation **saturates** into a limit cycle.
- Every design question that matters (amplitude, delivered power) is asked **here**, in the saturated regime.

Even without a load, the quantitative gap is about 25%



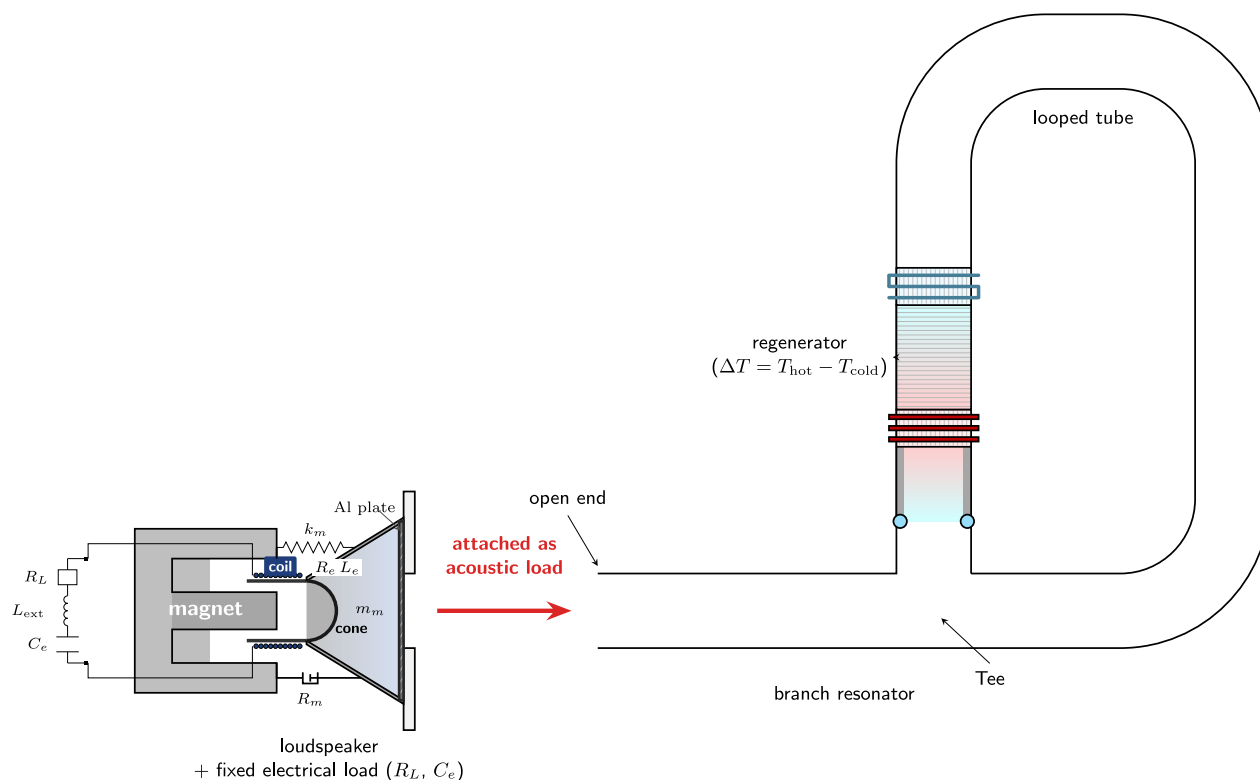
Redrawn after Swift, *Thermoacoustics* (2017), Fig. 7.1. Points: measured. Line: DeltaEC.



The textbook and the engine behind its data: the standing-wave engine of the Tektronix heat-driven refrigerator, 3 MPa helium, 370 Hz. Drawing redrawn after Swift, *Thermoacoustics* (2017), Fig. 1.9. Godshalk et al., *Adv. Cryog. Eng.* 41 (1996).

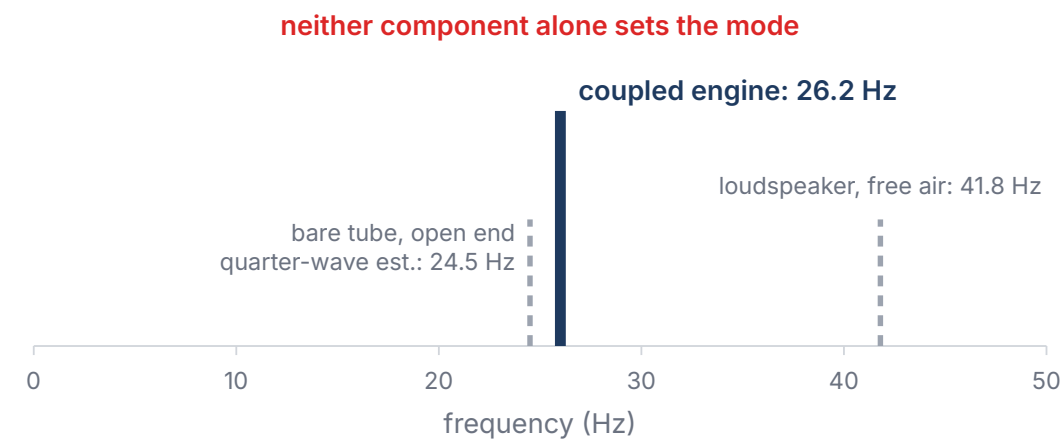
- Linear Rott theory asks the right question at **onset**. It is silent about amplitude.
- Quasi-linear extensions carry it into the saturated regime.
- On the textbook's own engine: $\approx 25\%$ in pressure, $\approx 200^\circ\text{C}$ in hot-end temperature.
- Quasi-linear design is still the field's **standard tool** today.

The working point belongs to the coupled system



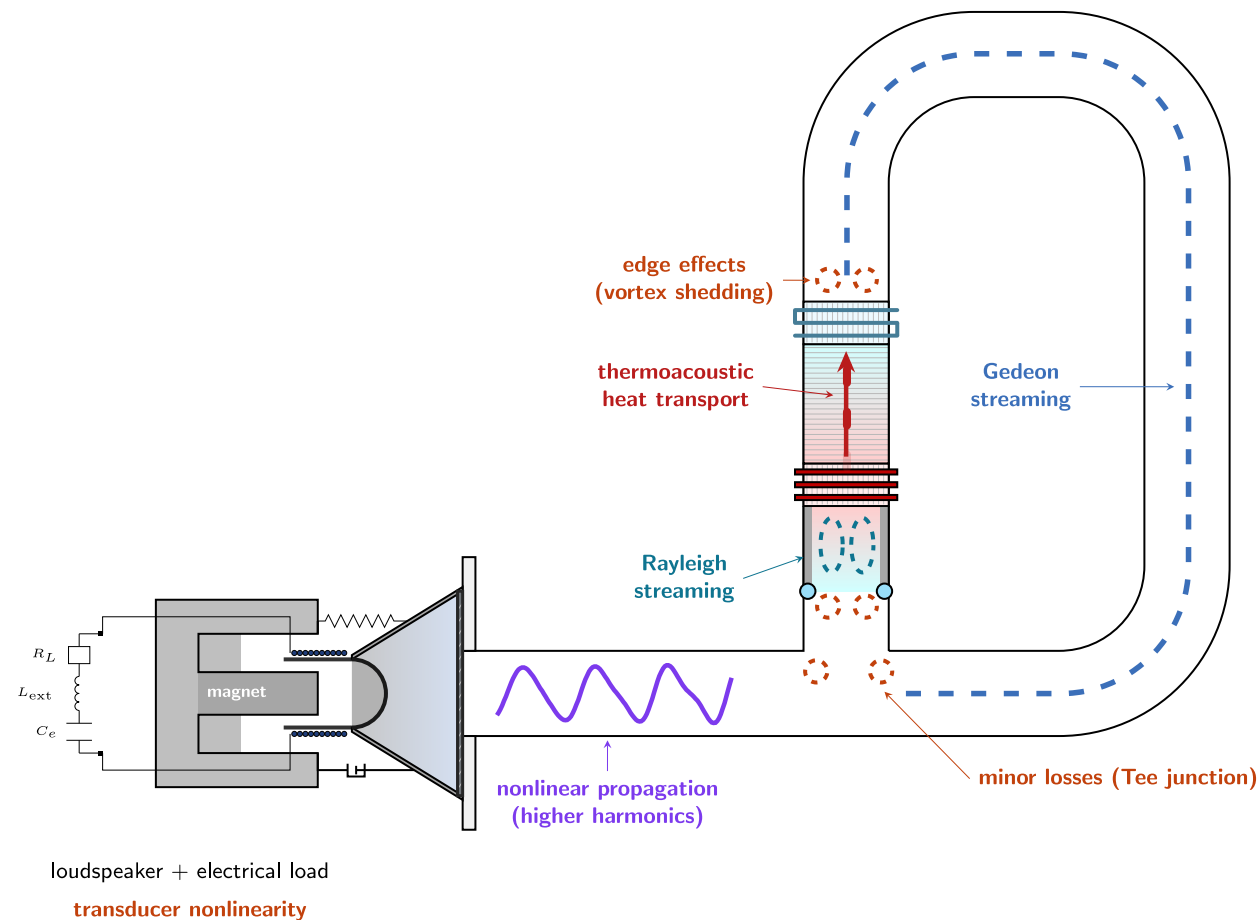
The engine and its loudspeaker load before attachment.

- Component-alone references: a **quarter-wave estimate** gives ≈ 24.5 Hz and the speaker alone resonates at **41.8 Hz**.
- Coupled, the engine runs at **26.2 Hz**. The speaker sits **nearly 40% below free-air resonance**, where the datasheet impedance no longer applies.
- CFD? A 26 Hz carrier, ≈ 0.8 s growth, hundreds of seconds of thermal ramp: not a design loop, and no design parameters at the end.



Working frequency: component-alone references vs the coupled system (measured).

The standard route: study each effect in isolation



Nonlinear processes on this engine's own geometry, after G. Penelet (2014). See Swift, *Thermoacoustics* (2017), ch. 7, "Beyond Rott".

isolate → correlate → superpose

(and hope the operating point transfers)

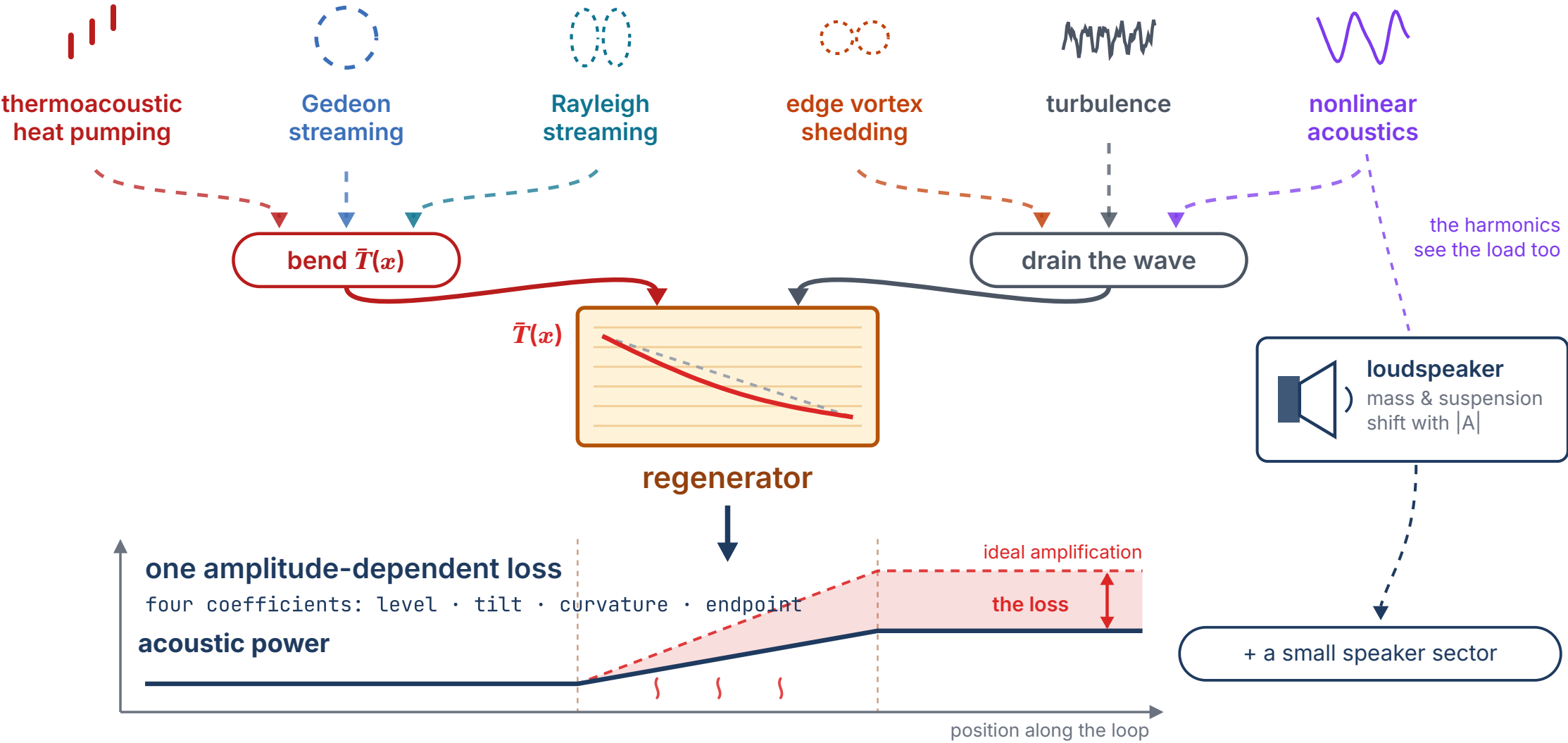
Nonlinear process	Understanding	Modelling	Impact on engines
Thermoacoustic heat pumping	good	good	significant
Nonlinear acoustics	good	good	minor
Streaming	good	poor	significant
Edge effects	good	poor	significant
Turbulence	poor	poor	unknown

Table redrawn from the survey compiled by G. Penelet at the 2nd Int. Workshop on Thermoacoustics, Sendai (2014).

The mechanisms with the biggest impact are the ones modelled worst.

A decade later, this table still holds.

They all land in the same place



Two questions, two answers, ten years apart

At the 2nd Int. Workshop on Thermoacoustics (Sendai 2014), G. Penelet closed his invited talk with two questions to the community, each arrow carrying his own answer. Ours, ten years on:

How to fit experiments and theory ? $\Rightarrow$ adjust any poorly known parameter ?

No. No knob per mechanism: their aggregate is **one regenerator loss shape, four coefficients, and the data prove they are identifiable.**

How to dissociate the role of each NL process ? $\Rightarrow$ study the transient regime

Yes. We did exactly that. Six measured transients, each from onset to saturated amplitude, **are** the data. An **adjoint** makes fitting them affordable.

Questions quoted verbatim from slide 23 of: G. Penelet, invited talk, 2nd Int. Workshop on Thermoacoustics, Sendai 2014.

And one standard throughout: **every approximation, including the ones inside our own optimizer, is checked against the genuine coupled system.**

1

Linear operator

geometry and load, measured,
not tuned

$$\mathbf{L} \mathbf{x} = i\sigma \mathbf{S} \mathbf{x}$$



2

Reduced envelope model

loss-shape structure from
physics

$$A_{n+1} = A_n e^{\theta_{\sigma}(\sigma_{\text{eff}} + i\omega_{\text{eff}})\Delta t}$$



3

Adjoint identification

on six measured transients

$$\partial \mathcal{J} / \partial \theta \leftarrow \mathbf{w}^H \delta \mathbf{L} \mathbf{v}$$



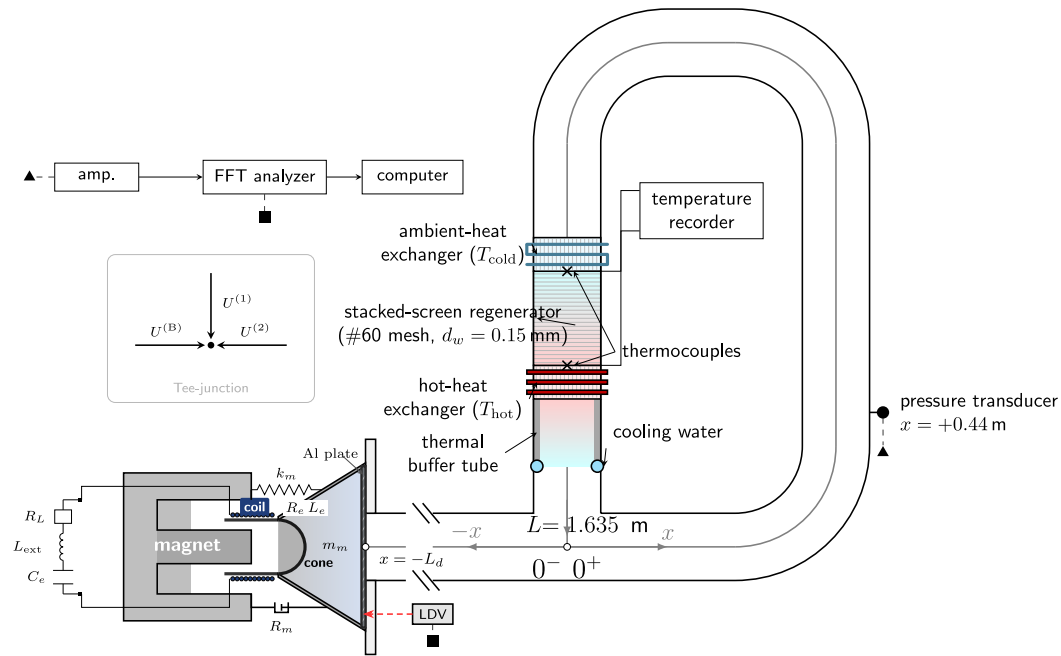
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Genuine-forward verification

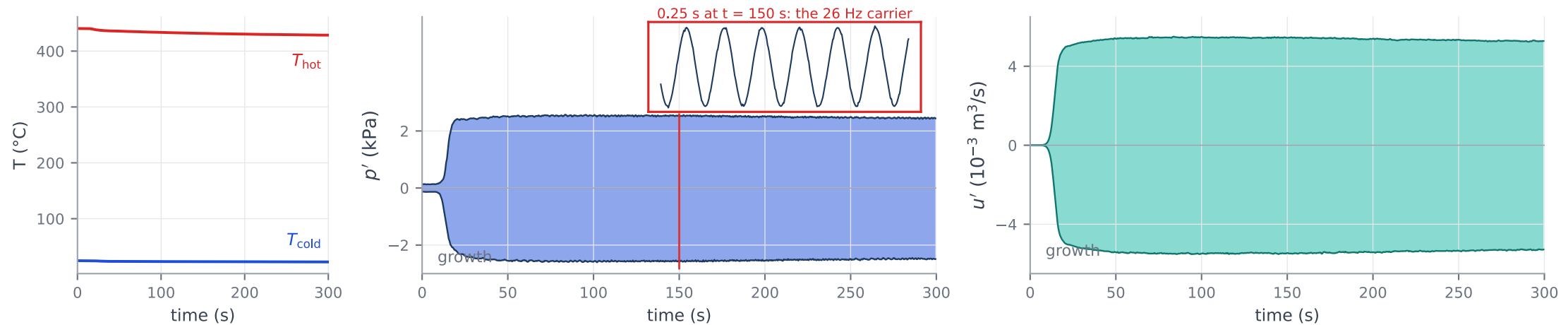
the only referee

$$(\mathbf{L} + \delta \mathbf{L}(\theta)) \mathbf{x} = i\tilde{\sigma} \mathbf{S} \mathbf{x}$$

The machine and the data



- **Working gas:** air at atmospheric pressure (101.3 kPa).
- Looped tube and branch resonator joined at a **Tee junction**. Loudspeaker termination with a fixed electrical load (500 Ω , 1 μ F).
- Measured channels: pressure at $x = 0.44$ m and loudspeaker **volume velocity** u' (laser vibrometry), both sampled at **3.2768 kHz**.
- **Six supercritical conditions**, $\Delta T = 365$ to 415 K, recorded as thermal transients.



Hottest condition ($\Delta T = 415$ K): thermal history, pressure p' , volume velocity u' . Bands: 0.5-s min to max, first 300 s. Inset: the 26 Hz carrier.

From Rott's equations to nodal blocks

Rott momentum and continuity (thermoacoustic production via $T_m(x)$):

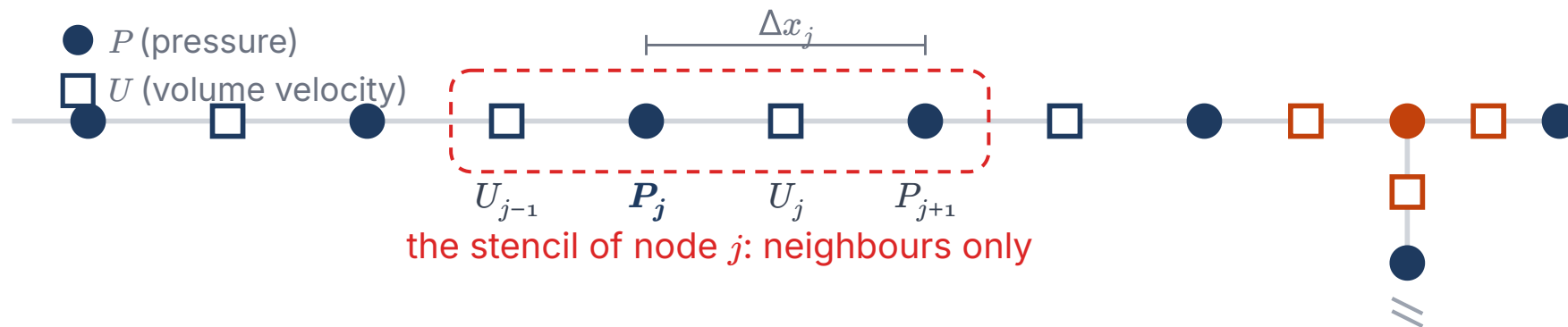
$$i\sigma U = \left[- (1 - \chi_\nu) \frac{A}{\rho_m} \frac{d}{dx} \right] P$$

$$i\sigma P = \left[\frac{\gamma P_m}{A[1 + (\gamma - 1)\chi_\alpha]} \left(\frac{\chi_\alpha - \chi_\nu}{(1 - \chi_\nu)(1 - \text{Pr})} \frac{1}{T_m} \frac{dT_m}{dx} - \frac{d}{dx} \right) \right] U$$

discretised at node j , on the staggered grid:

$$\left(\frac{\mathcal{G}_j}{2} + \mathcal{K}_j \right) U_{j-1} + \left(\frac{\mathcal{G}_j}{2} - \mathcal{K}_j \right) U_j = i\sigma P_j$$

$$\mathcal{M}_j^{-1} P_j - \mathcal{M}_j^{-1} P_{j+1} = i\sigma U_j$$



the stencil of node j : neighbours only

discrete coefficients (mobility · stiffness · thermoacoustic gain):

$$\mathcal{M}_j^{-1} = (1 - \chi_{\nu,j}) \frac{A_j}{\rho_{m,j} \Delta x_j}$$

$$\mathcal{K}_j = \frac{\gamma P_m}{A_j [1 + (\gamma - 1)\chi_{\alpha,j}] \Delta x_j}$$

$$\mathcal{G}_j = \mathcal{K}_j \frac{\chi_{\alpha,j} - \chi_{\nu,j}}{(1 - \chi_{\nu,j})(1 - \text{Pr})} \frac{T_{m,j+1} - T_{m,j-1}}{2 T_{m,j}}$$

P, U : complex pressure and volume-velocity amplitudes · $i\sigma = \sigma_r$

+ $i\omega$: the eigenvalue (growth rate + frequency) · χ_ν, χ_α :

viscous and thermal Rott functions · A : duct area · $\rho_m, P_m, T_m(x)$:

mean density, pressure, temperature · γ : specific-heat ratio · Pr :

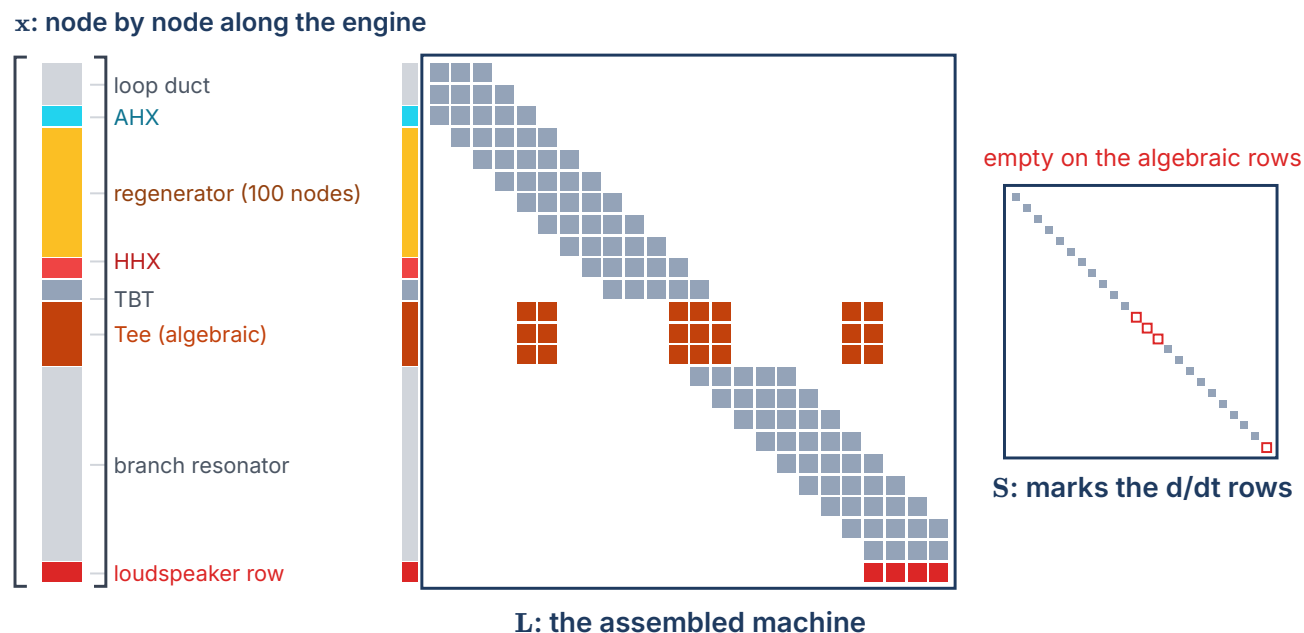
Prandtl number · Δx_j : grid step at node j

Tee: three legs meet, $\Sigma U = 0$

The differential equations have become linear relations among neighbouring P_j and U_j : rows of a matrix, one pair per node.

Everything in $\mathcal{M}_j, \mathcal{K}_j, \mathcal{G}_j$ is geometry, gas properties and the measured $T_m(x)$.

Assembling the machine: L and S



Structure schematic: 24 rows stand in for the 145 nodes.

$$\mathbf{L} \mathbf{x} = i\sigma \mathbf{S} \mathbf{x}$$

One eigenproblem per ΔT . No free parameters anywhere.

- Stack the unknowns **along the acoustic path**. The

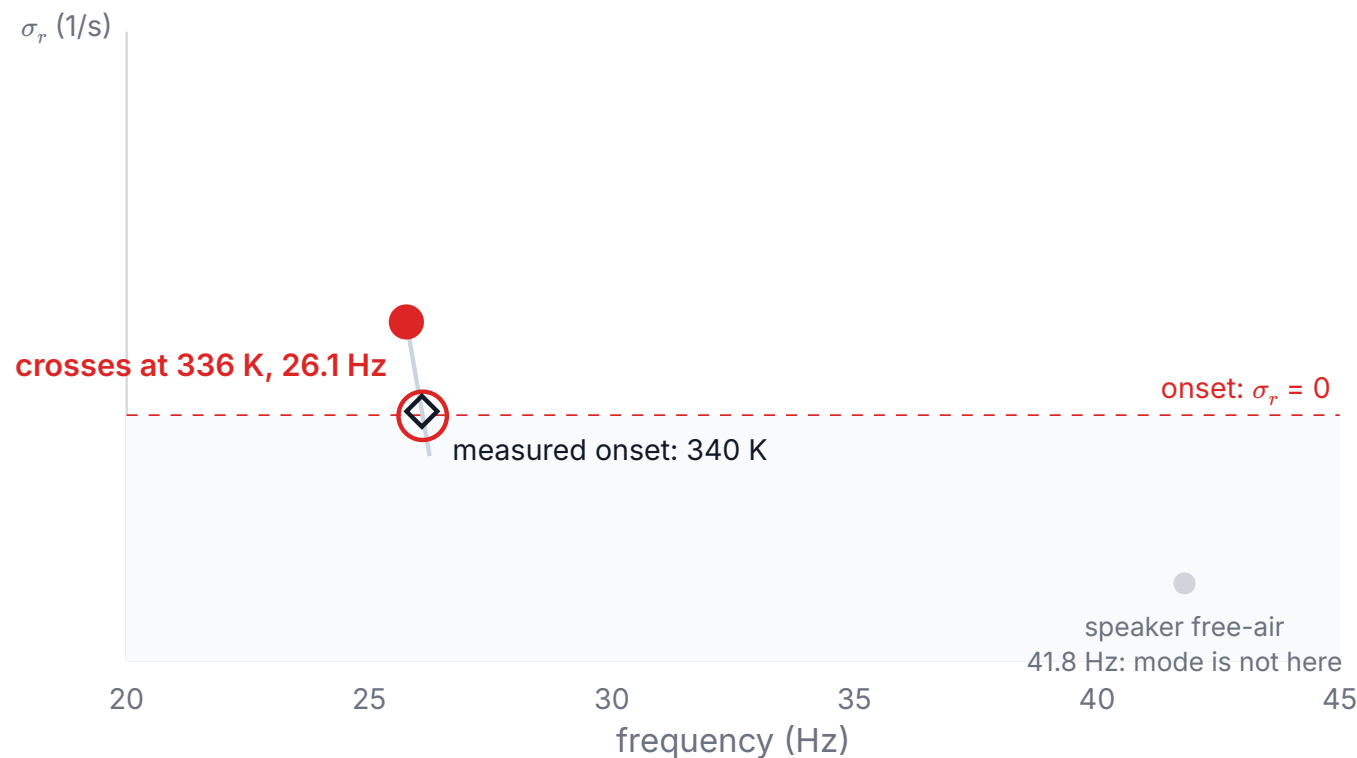
vector is the engine's anatomy: $\mathbf{x} = \begin{bmatrix} P_1 \\ U_1 \\ \vdots \\ U_N \end{bmatrix}$

- L assembles the machine**: the two Rott rows at every node, with each component's geometry and $T_m(x)$:

$$\begin{aligned} \left(\frac{g_j}{2} + \mathcal{K}_j\right)U_{j-1} + \left(\frac{g_j}{2} - \mathcal{K}_j\right)U_j &= i\sigma P_j \\ \mathcal{M}_j^{-1}P_j - \mathcal{M}_j^{-1}P_{j+1} &= i\sigma U_j \end{aligned}$$

- One boundary row from the measured speaker: $P_N = Z_{\text{sp}}U_N$
- Three Tee rows: equal pressures, $\Sigma U^{(\ell)} = 0$
- S selects** the rows that carry d/dt: identity on the $2N-4$ differential rows, **empty on the 4 algebraic rows**.

Backbone: the linear operator knows the coupled mode



$$\mathbf{L}(\Delta T) \mathbf{x} = i\sigma \mathbf{S} \mathbf{x}, \quad i\sigma = \sigma_r + i\omega$$

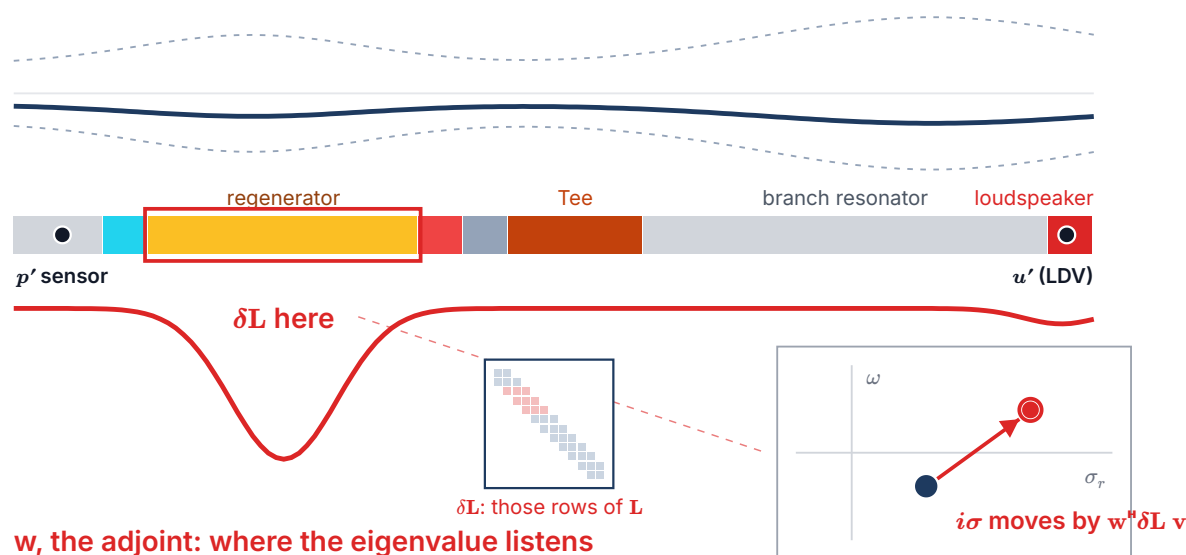
- Rott equations + Tee junction + loudspeaker impedance boundary, discretised to a generalized eigenproblem.
- **Zero calibrated parameters.** Onset: **336 K** computed vs 340 K measured. Frequency: **26.1 Hz** vs 26.2 Hz.
- The load's near-40% frequency shift is **not an input but a prediction.**

$\Delta T = 420 \text{ K} \cdot f = 25.8 \text{ Hz} \cdot \sigma_r = 1.34 \text{ 1/s} \cdot \text{growing}$

Dominant eigenvalue vs heating. Trajectory schematic between computed points, onset and frequency as computed.

The mode and its adjoint

$\mathbf{v}$, the mode: the computed oscillation pattern of the whole machine



$\mathbf{w}$, the adjoint: where the eigenvalue listens

Adjoint-based sensitivity: Magri and Juniper (2013), Juniper and Sujith (2018).

$$\frac{dA}{dt} = [i\sigma + \mathbf{w}^H \delta \mathbf{L} \mathbf{v}] A \quad (\text{with } \mathbf{w}^H \mathbf{S} \mathbf{v} = 1)$$

The same contraction on the perturbed dynamics: the **amplitude equation**, the reduced model of the next slide.

$$\mathbf{L} \mathbf{v} = i\sigma \mathbf{S} \mathbf{v}$$

- **Right eigenvector $\mathbf{v}$** : the dominant solution $\mathbf{x}$, now named. The **computed** mode shape $P(x)$, $U(x)$, sampled by the sensors at just two points.

$$\mathbf{w}^H \mathbf{L} = i\sigma \mathbf{w}^H \mathbf{S}$$

- **Left eigenvector $\mathbf{w}$** : the **adjoint**: where the eigenvalue listens.

perturb the machine, $\mathbf{L} \rightarrow \mathbf{L} + \delta \mathbf{L}$, and keep first order:

$$(\mathbf{L} + \delta \mathbf{L})(\mathbf{v} + \delta \mathbf{v}) = (i\sigma + \delta(i\sigma)) \mathbf{S}(\mathbf{v} + \delta \mathbf{v})$$

project with $\mathbf{w}^H$, the unknown $\delta \mathbf{v}$ cancels:

$$\mathbf{w}^H \mathbf{L} \delta \mathbf{v} - i\sigma \mathbf{w}^H \mathbf{S} \delta \mathbf{v} + \mathbf{w}^H \delta \mathbf{L} \mathbf{v} = \delta(i\sigma) \mathbf{w}^H \mathbf{S} \mathbf{v}$$

the change, felt by $\mathbf{v}$, heard by $\mathbf{w}$

$$\Rightarrow \delta(i\sigma) = \frac{\overbrace{\mathbf{w}^H \delta \mathbf{L} \mathbf{v}}}{\underbrace{\mathbf{w}^H \mathbf{S} \mathbf{v}}_{\text{normalisation}}}$$

- $\delta \mathbf{v}$ **drops out**: perturb *anything*, the shift follows with **no re-solve**.
- $\delta \mathbf{L}$ is **sandwiched between $\mathbf{w}$ and $\mathbf{v}$** : it matters only where the mode moves *and* the eigenvalue listens. The **overlap peaks at the regenerator**.

One regenerator loss shape

The unknowns first, the promised four plus a growth time-scale θ_σ and a small speaker sector:

$$\theta = (\underbrace{\theta_{\text{reg}}}_{\text{level}}, \underbrace{\theta_{\text{tilt}}}_{\text{tilt}}, \underbrace{\theta_{\text{curve}}}_{\text{curvature}}, \underbrace{\theta_{\text{end}}}_{\text{endpoint}}, \underbrace{\theta_{\text{spk}}}_{\text{speaker sector}}, \dots)$$

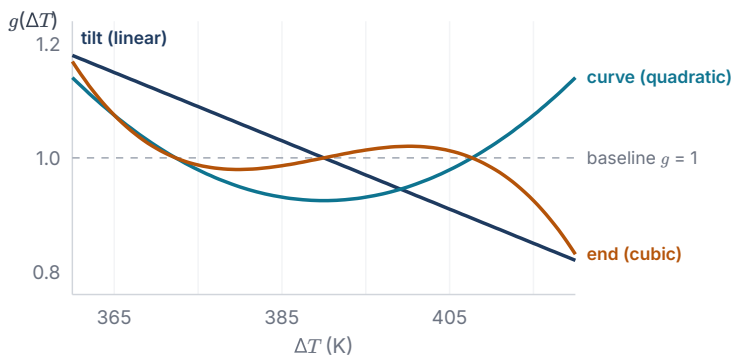
Project onto the dominant mode, then advance the complex envelope:

$$A_{n+1} = A_n \exp[\theta_\sigma (\sigma_{\text{eff},n} + i \omega_{\text{eff},n}) \Delta t_n]$$

$$\sigma_{\text{eff}} = \sigma_r(\Delta T) - \sigma_{\text{reg}}^{\text{nl}} - \sigma_{\text{spk}}^{\text{nl}}$$

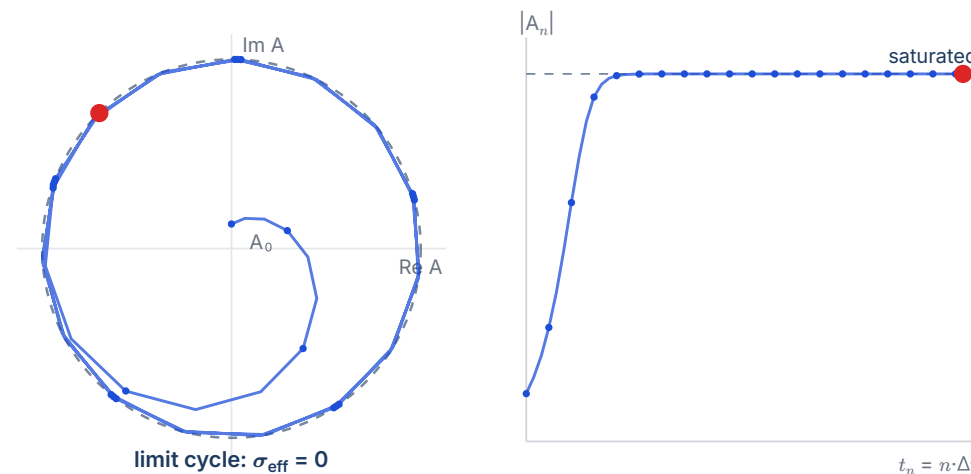
$$\sigma_{\text{reg},n}^{\text{nl}} = \underbrace{\theta_{\text{reg}}}_{\text{loss level}} \underbrace{g(\Delta T_n)}_{\text{shape along the branch}} \underbrace{\Sigma_{\text{reg}}(Re_{h,n})}_{\text{grows with } |A_n|}$$

$$g(\Delta T) = 1 + \theta_{\text{tilt}} \phi_{\text{tilt}} + \theta_{\text{curve}} \phi_{\text{curve}} + \theta_{\text{end}} \phi_{\text{end}}$$



The three admissible deformations of the loss, each drawn at an illustrative size.

Fixed shapes, not fitted: $\phi_{\text{tilt}} = s$, $\phi_{\text{curve}} = s^2 - \frac{1}{2}$, $\phi_{\text{end}} = s(s^2 - \frac{1}{2})$, with $s = (390 - \Delta T)/25$



The envelope, updated step by step in time: each $\Delta t_n = 0.5$ s multiplies A_n by $\exp[\theta_\sigma (\sigma_{\text{eff}} + i \omega_{\text{eff}}) \Delta t_n]$, a stretch and a rotation. Each A_n is demodulated from the 3.3 kHz records. As $|A|$ grows, σ_{eff} shrinks and the spiral lands on the limit cycle.

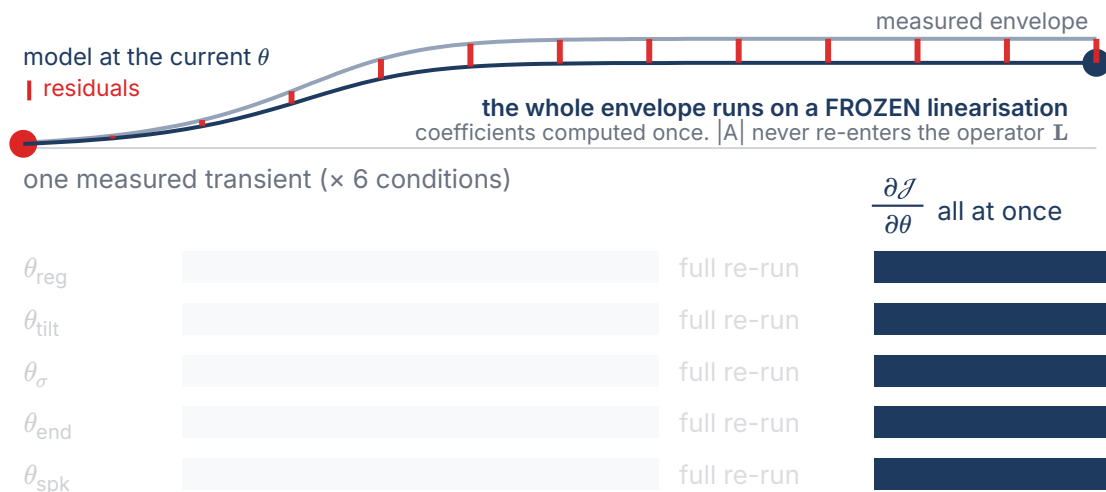
- **Structure from physics:** the loss is the aggregate of the un-modelled mechanisms. It lives in the regenerator and grows with velocity. The small speaker sector is identified too (backup).
- **Numbers left to the data:** one set of θ for the whole branch, not one fit per condition. **Identifying θ is the rest of the talk.**

$$\Sigma_{\text{reg}} = 0.8 \nu Re_h / d_h^2, \quad Re_h = d_h |V_{\text{reg}}| / \nu$$

Tanaka's oscillatory screen friction, geometric $d_h = 0.41$ mm. Derived, not fitted.

A product because the friction's spatial pattern is fixed: its projection is one constant, folded into θ_{reg} (backup).

The gradient engine: one sweep, every parameter



adjoint: full gradient, every parameter, one sweep

One forward run, one backward sweep: every residual collects into the full gradient $\partial \mathcal{J} / \partial \theta$.

$$\mathcal{J}(\theta) = \frac{1}{N_c} \sum_{c=1}^6 \frac{1}{2} \sum_{\ell=1}^8 w_{\ell} (R_{\ell}^{(c)})^2$$

$$R_{\ell}^{(c)} = \underbrace{\text{model}_{\ell}(\theta)}_{\text{reduced forward}} - \underbrace{\text{data}_{\ell}^{(c)}}_{\text{measured envelopes}}$$

- One backward sweep per iteration gives the **full gradient**, however many parameters there are.
- It also **ranks sensitivities** and **warm-starts** the search.

$\mathcal{J}$: the transient misfit · θ : the coefficient vector (loss shape + speaker sector) · $N_c = 6$ operating conditions, indexed by $c \cdot \ell = 1 \dots 8$ residual channels, fixed weights w_{ℓ} :

R_p	pressure envelope	1.0	R_g	growth rate	0.5
R_u	u' envelope	0.75	R_f	carrier frequency	0.15
$R_{T,a}$	mobility amplitude	1.25	R_d	detuning	0.5
$R_{T,\varphi}$	mobility phase	0.75	R_t	saturated tail	1.0

1

Six measured transients
envelopes, mobility, frequency



2

Residuals
model(θ) minus data



3

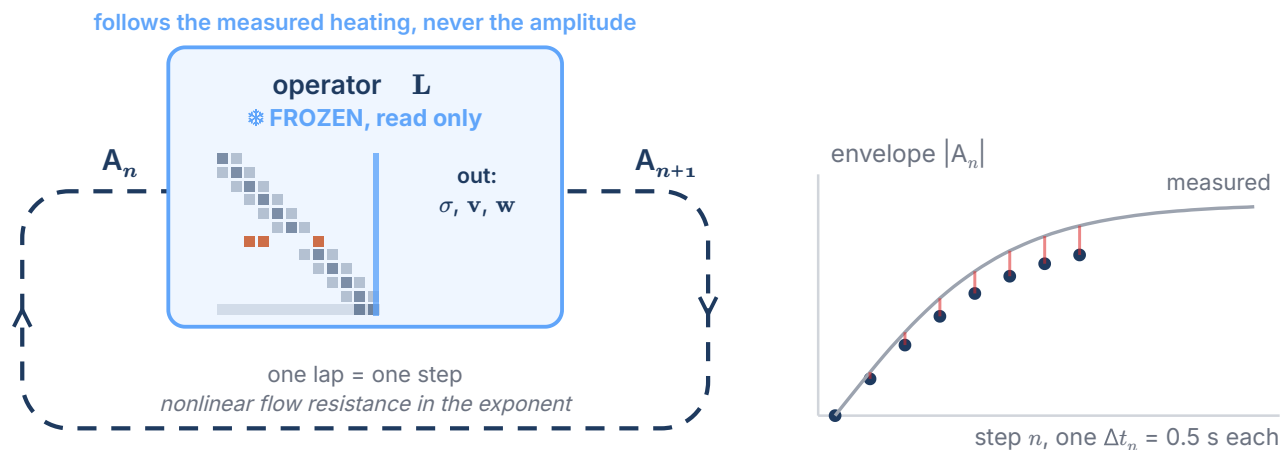
One backward sweep
the full gradient of $\mathcal{J}$



4

Update θ , repeat
until the transients are fit

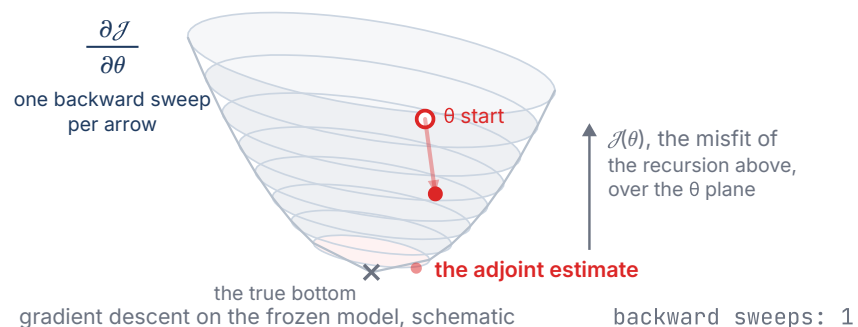
One catch: the adjoint runs on a frozen linearisation



$$A_{n+1} = A_n \exp \left[\theta_\sigma (\sigma_{\text{eff},n} + i \omega_{\text{eff},n}) \Delta t_n \right] = \text{one lap of the loop}$$

The envelope is a recursion through the operator, and the box is the sparse $\mathbf{L}$ itself: its coefficients follow the measured heating, and the amplitude is never written back.

speed $\times 1.0$ sweep 1 / 7 click the staircase to jump



Freezing is an *isolation*:

$$\frac{dA}{dt} = \theta_\sigma \left[\underbrace{i\sigma(\Delta T_n)}_{\text{frozen}} + \underbrace{\mathbf{w}^H}_{\text{frozen}} \delta \mathbf{L} \underbrace{\mathbf{v}}_{\text{frozen}} \right] \underbrace{A}_{\text{evolves}}$$

Measured values enter through $\mathbf{L}$: the temperature history $\Delta T(t)$ and the bench Z_{sp} . The frozen $i\sigma$ tracks the measured heating, never the amplitude.

The $|A|$ scalars inside $\delta \mathbf{L}$ stay live, so the envelope still saturates. The operator behind $\sigma, \mathbf{v}, \mathbf{w}$ is **never re-solved as the amplitude grows**.

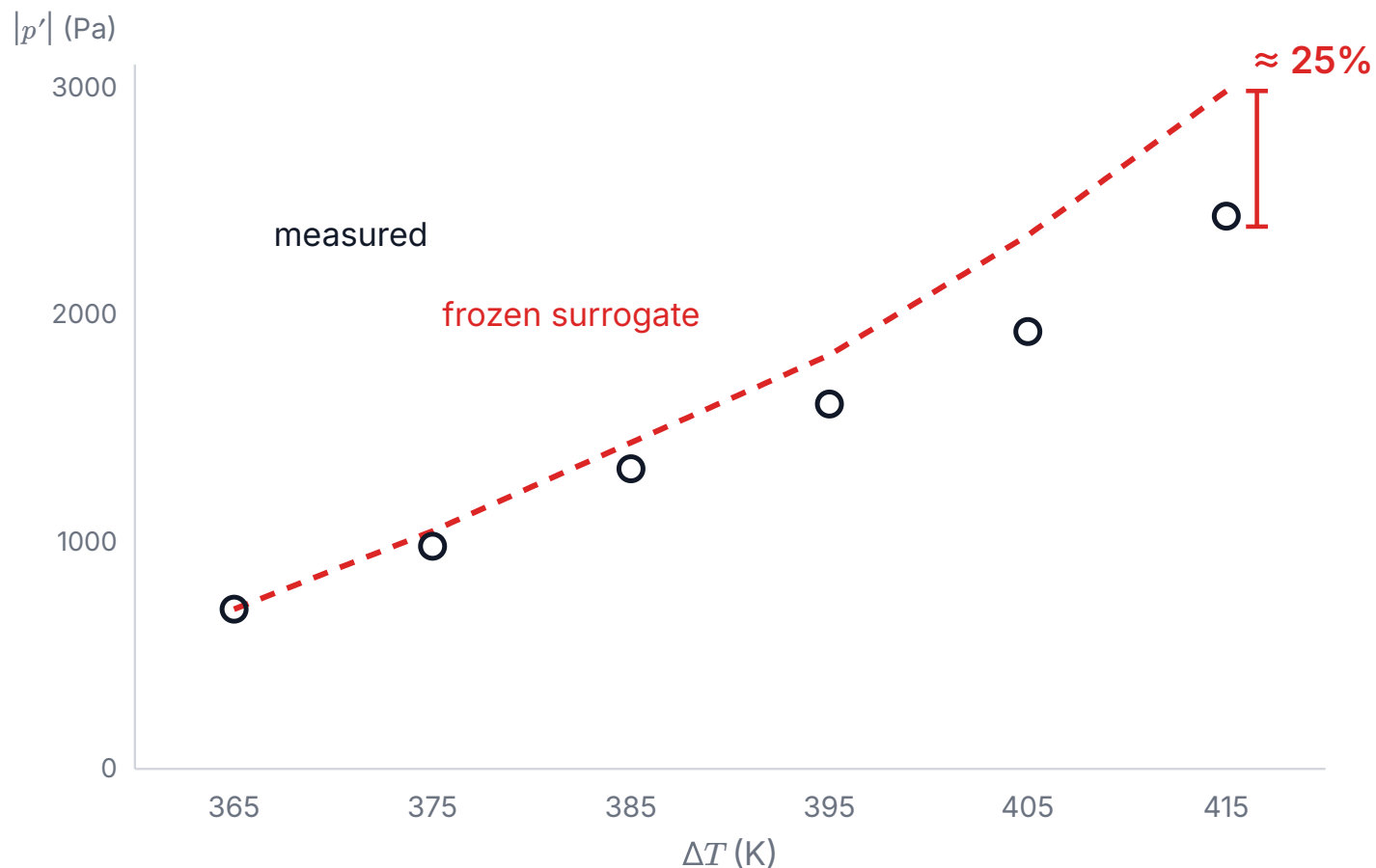
That live scalar is the Tanaka screen friction, via the nonlinear flow resistance:

$$f_h = \frac{175}{Re_h} + 1.60, \quad R_\nu = \frac{0.8 \rho \nu}{A_r d_h^2} Re_h - \frac{\omega \rho}{A_r} \text{Im} \left[\frac{1}{1 - \chi_\nu} \right]$$

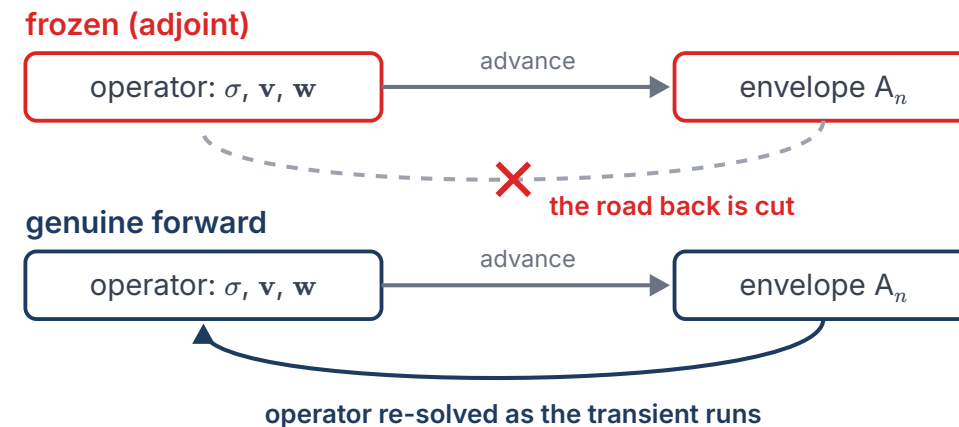
slope from Tanaka, the $|V| = 0$ limit from Ueda. Obayashi, Hsu and Biwa, TEION KOGAKU **47**, 562 (2012).

So how good is the frozen surrogate's branch?

The price of freezing



Measured points and the frozen-surrogate branch. Bias schematic, capped at the observed $\approx 25\%$.

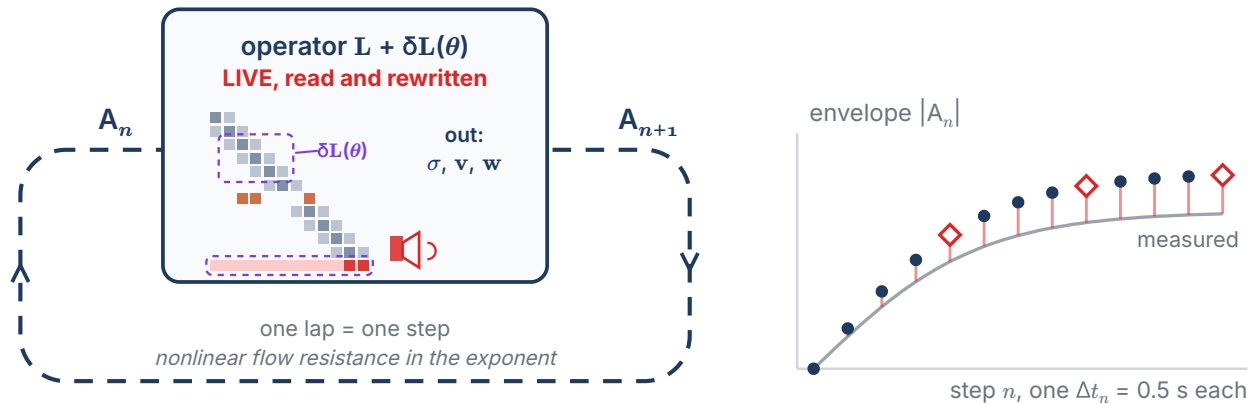


Close θ , wrong messenger. The descent lands a warm start close to the answer. But this forward has the road back cut, and its branch overshoots the measurements by up to **25%**.

So let the operator live: **re-solve it as the transient runs.**

The genuine forward, step by step

rebuilt at the measured temperatures every $m = 20$ steps, then re-solved

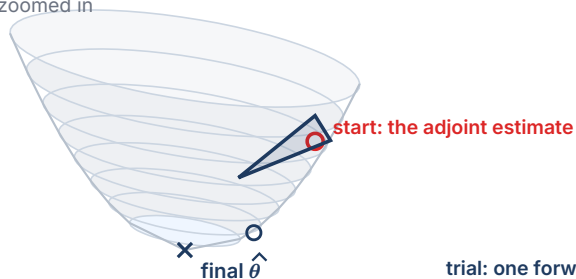


$$A_{n+1} = A_n \exp \left[\theta_\sigma (\sigma_{\text{eff},n} + i \omega_{\text{eff},n}) \Delta t_n \right] = \text{one lap, both forwards}$$

The same recursion: every lap reads the sparse $\mathbf{L} + \delta\mathbf{L}(\theta)$ in the box, and every m -th lap **rebuilds** it at the measured temperatures and re-solves, refreshing the projections $\sigma, \mathbf{v}, \mathbf{w}$ that the exponent uses.

speed $\times 1.0$ trial 0 / 9 click the staircase to jump

the last stretch, zoomed in



no gradients, just cost values

trial: one forward run
forward runs: 4

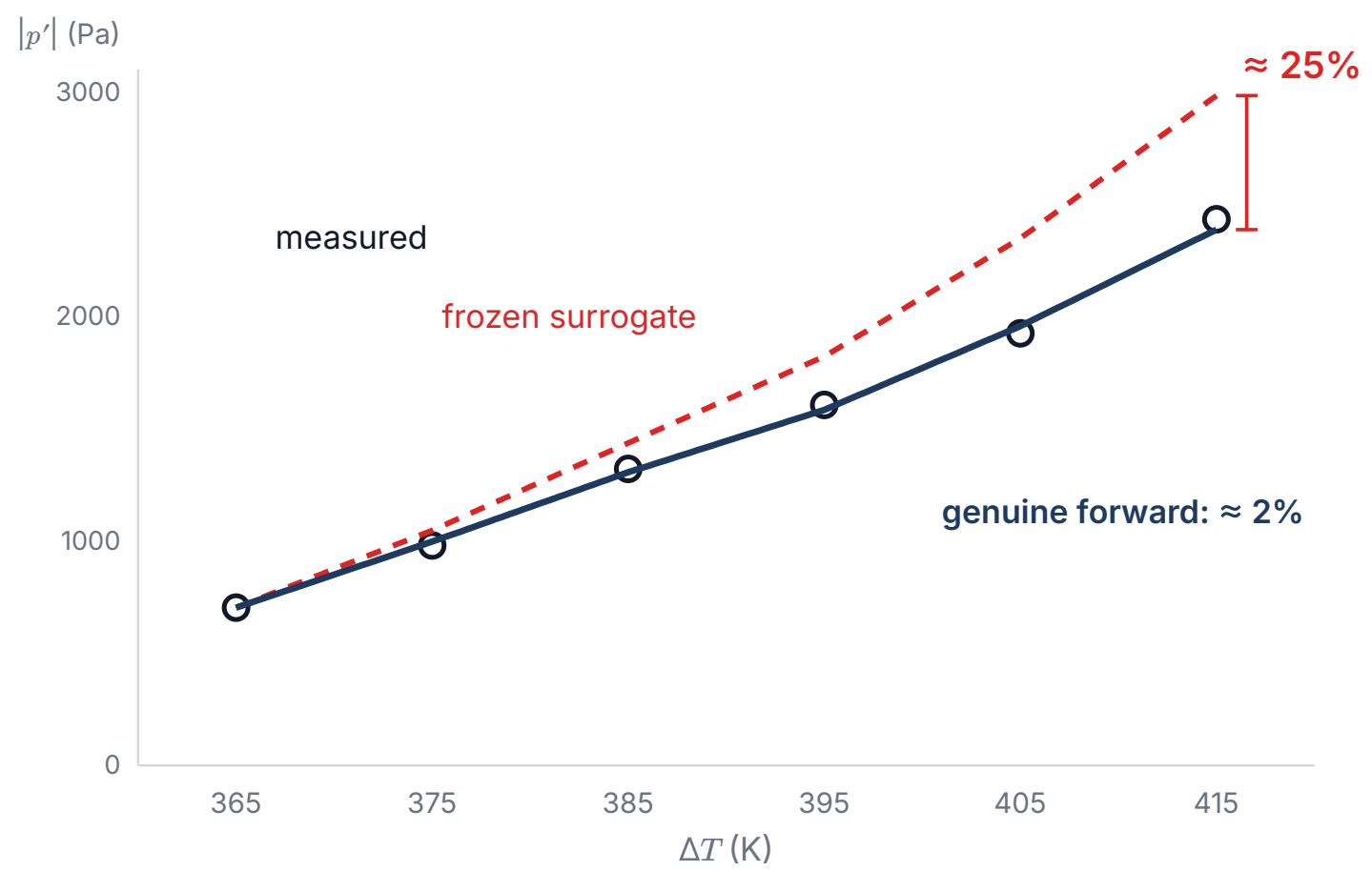
The same equation, now **live**:

$$\frac{dA}{dt} = \theta_\sigma \left[\underbrace{i\sigma(\Delta T_n, \theta)}_{\text{re-solved}} + \underbrace{\mathbf{w}^H}_{\text{refreshed}} \underbrace{\delta\mathbf{L}_n(\theta)}_{\text{refreshed}} \underbrace{\mathbf{v}}_{\text{refreshed}} \right] \underbrace{A}_{\text{evolves}}$$

- Every $m = 20$ steps: **rebuild** $\mathbf{L} + \delta\mathbf{L}(\theta)$ at the measured $T_{\text{hot}}(t), T_{\text{cold}}(t)$, **re-solve** the eigenproblem, track the eigenvector.
- That refreshes every underbraced symbol, the growth rate, the mode shape, the ratios behind Re_h . The frozen surrogate replays them computed once. Here they **track the running engine**.

Gradient-free Nelder–Mead: each trial point costs one complete forward run, the whole staircase above. Warm-started close by the adjoint, a few dozen runs to the $\approx 2\%$ branch. Refresh interval $m = 20$ steps.

The genuine forward, and the close



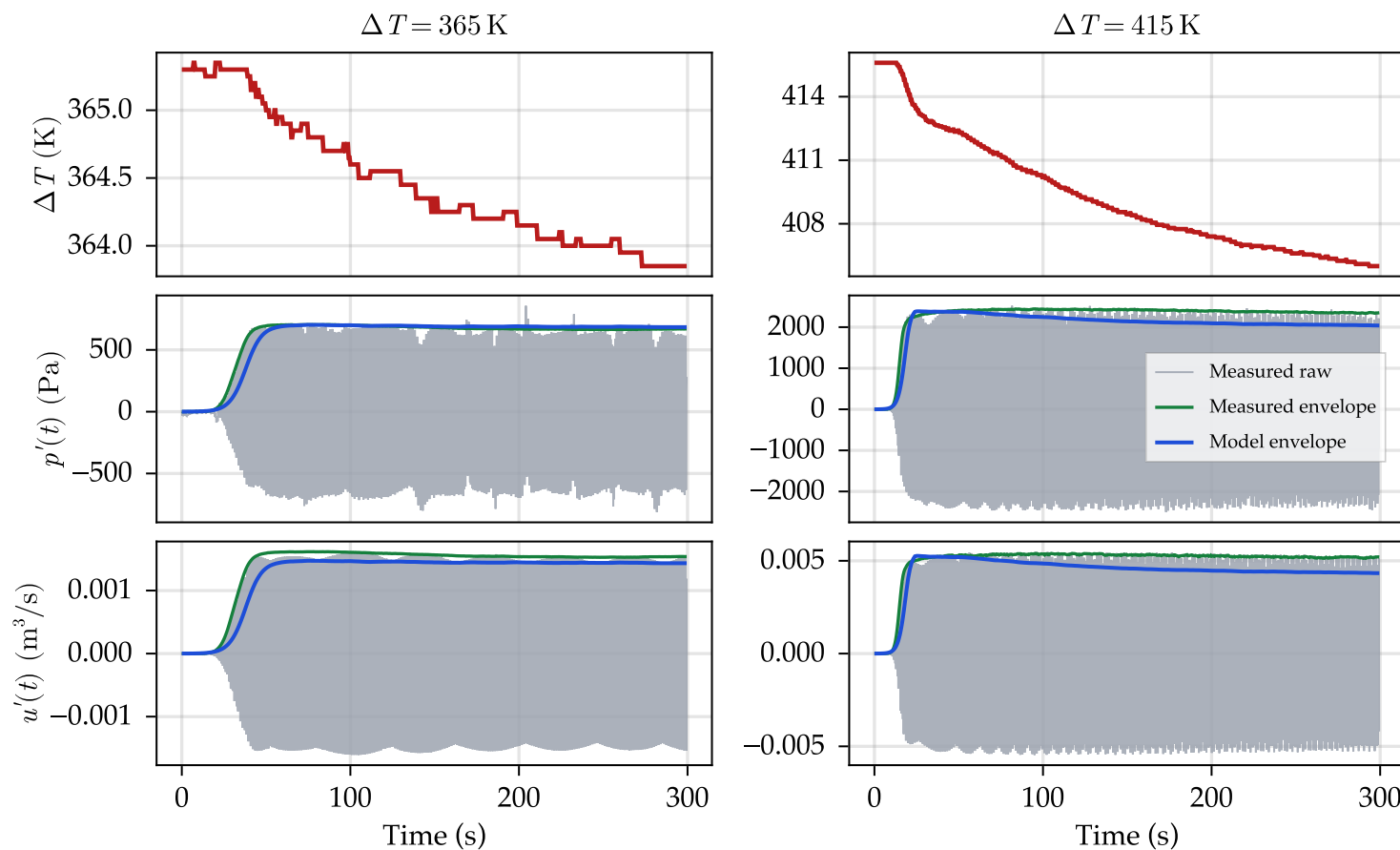
Measured points and genuine-forward branch as published. Frozen-surrogate bias schematic, capped at the observed $\approx 25\%$.

	frozen (adjoint)	genuine forward
branch error	$\approx 25\%$	$\approx 2\%$
gradient	1 backward sweep	forward evaluations
identification	gradient descent, adjoint sweep	gradient-free polish, warm-started
role	gradient · ranking · warm start	calibration · every reported number

ΔT (K)	$ p' $ error (%)	$ u' $ error (%)
365	-0.08	-8.7
375	+1.70	-6.2
385	-1.18	-7.8
395	-1.39	-7.5
405	+1.62	-3.6
415	-1.91	-1.7
mean $ \cdot $	1.31	5.92

- p' , the calibrated channel: mean **1.31%**, every case within 2%.
- u' , never calibrated: 5.92%, a constant offset. **The independent check.**

Where the model ends



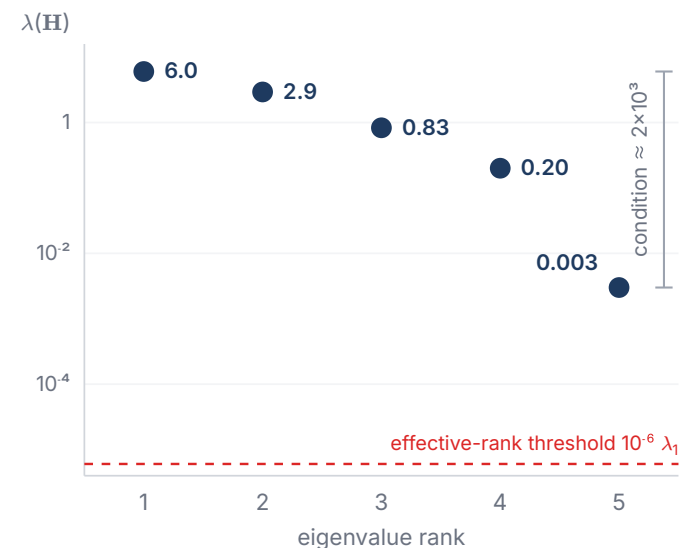
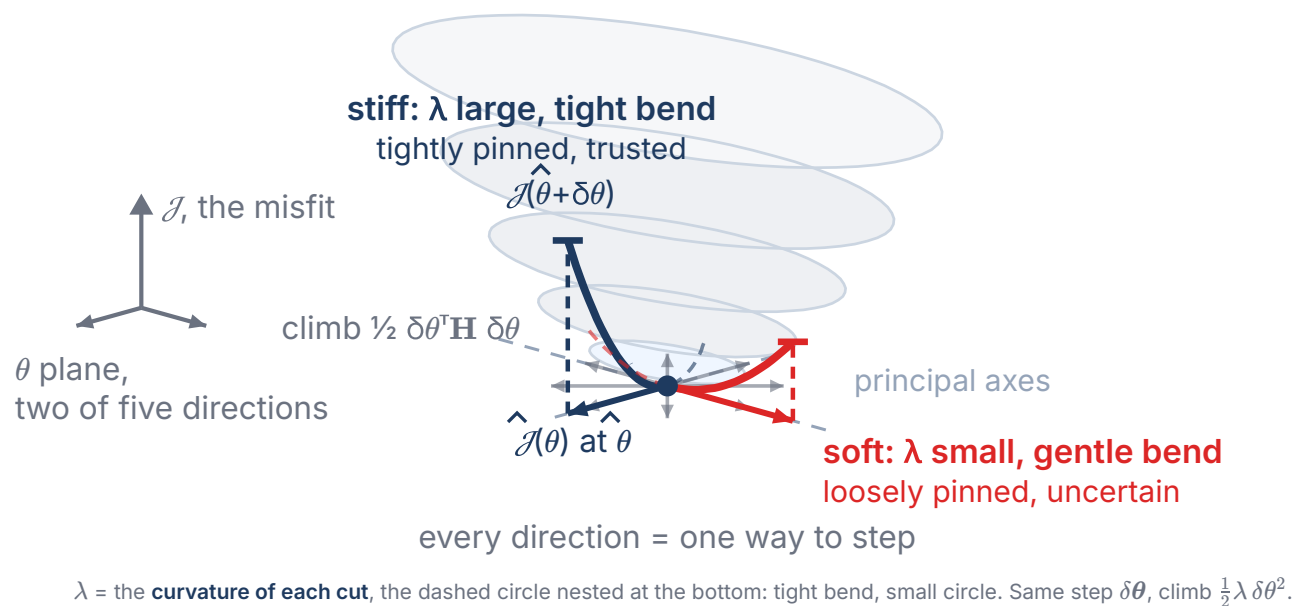
- Saturated amplitudes match, but the growth phases **do not quite line up**, the mismatch largest at $\Delta T = 365$ K.
- Residual power concentrates **below 0.1 Hz**: orders of magnitude slower than the 26 Hz carrier and the ≈ 0.8 s growth. Instantaneous-amplitude corrections cannot carry a drift this slow.

The residual **names the missing state**: a regenerator temperature $\bar{T}_{\text{reg}}(t)$ with its own relaxation.

One extra state. The adjoint machinery extends unchanged.

Identification, not fitting

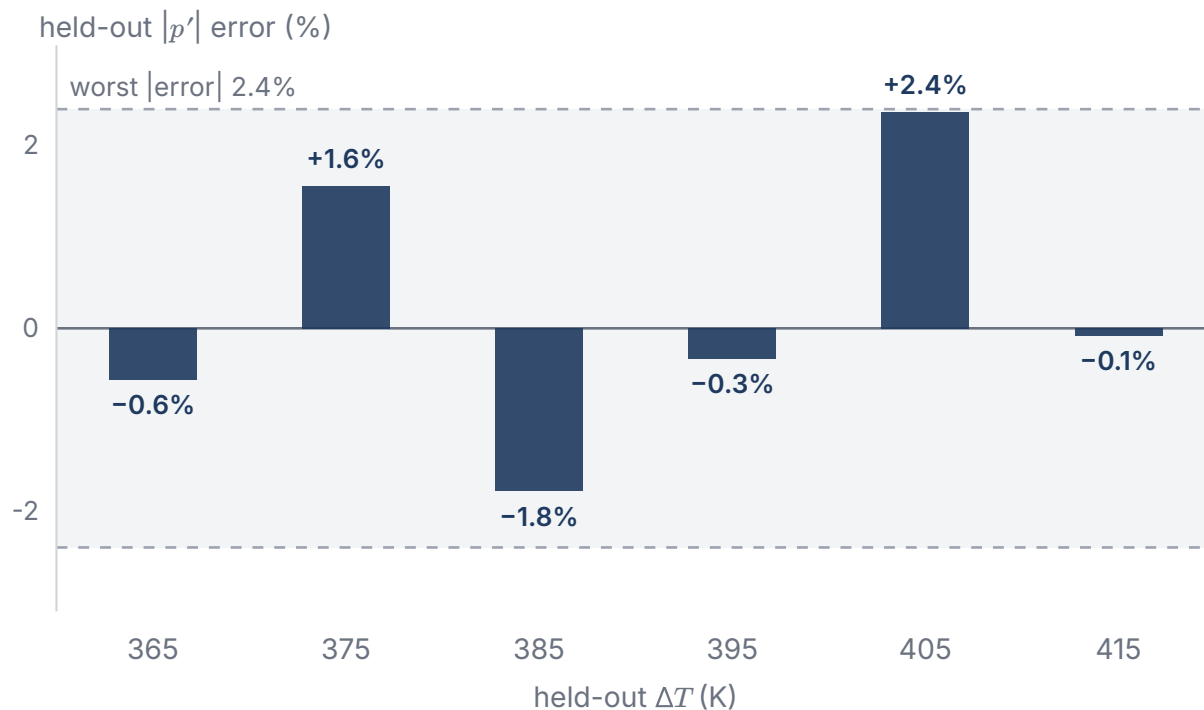
$$\mathcal{J}(\hat{\theta} + \delta\theta) \approx \mathcal{J}(\hat{\theta}) + \frac{1}{2} \delta\theta^T \mathbf{H} \delta\theta$$



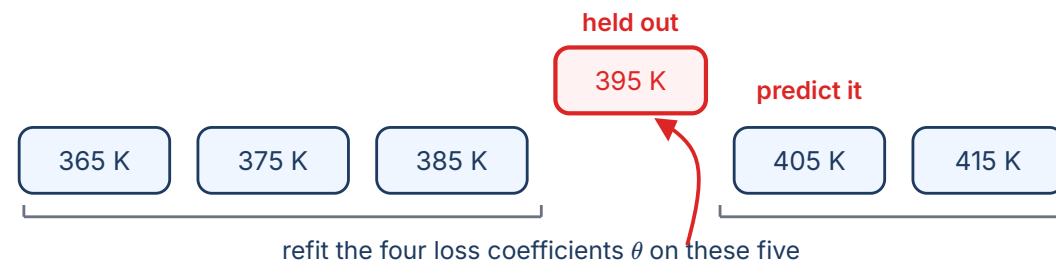
The five climbs $\lambda(H)$, evaluated on the genuine forward.

- The eigenvalues $\lambda(H)$: the curvature of each principal cut. 0.003 to 6.0, **full effective rank**. Every cut bends upward, none is flat.
- This is the answer to "adjust any poorly known parameter?": the curvature of the data decides, not the modeller.**

The loss shape transfers



Held-out pressure-branch error per fold, refit on the remaining five conditions.



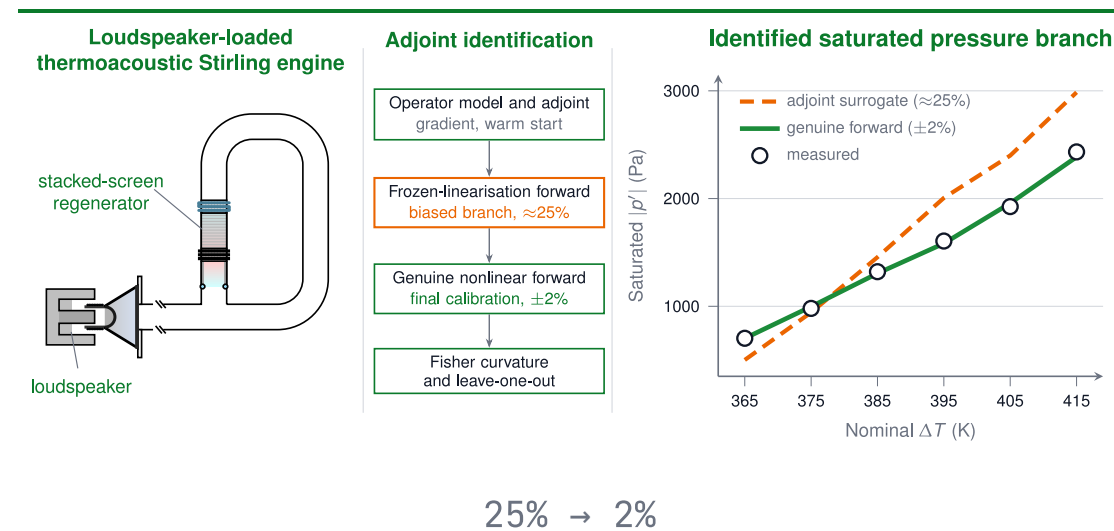
six folds, each condition takes a turn

- Held-out pressure reproduced to a **median 1.1%**, **worst 2.4%**, against the in-sample mean 1.3% and worst 1.9%.
- The refit loss shape barely moves between folds, with no outlier fold.

The loss shape is a property of the machine, not of the data set.

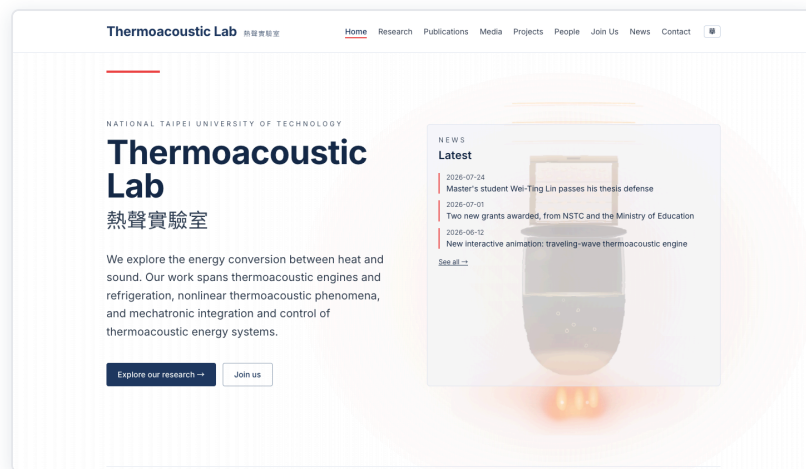
Conclusion

- **Operator:** onset 336 K, frequency 26 Hz, with zero calibration. The load's mode shift is a prediction.
- **Loss shape:** four coefficients reproduce the saturated pressure branch to $\approx 2\%$, with u' tracked as an independent check.
- **Standard:** the frozen adjoint supplies gradients. The genuine forward supplies every reported number.
- **Reliability:** identifiability (full rank) and leave-one-out transfer ($\leq 2.4\%$) instead of adjusted knobs.



Journal version: Hsu and Lai, submitted to J. Sound Vib. (2026) · Data: doi:10.17632/c3fkjxryt.1

Thank you. Questions?

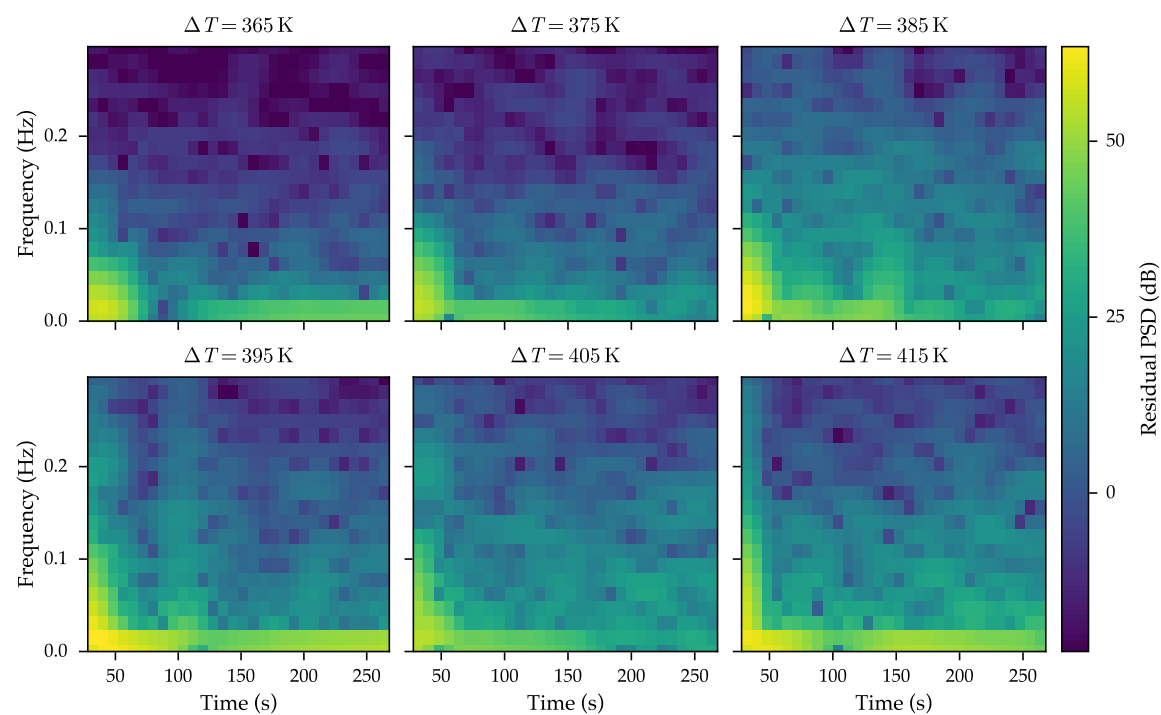


Thermoacoustic Lab · Taipei Tech
thermoacoustictw.org/en

For the slides, email bookhsu@ntut.edu.tw

BACKUP

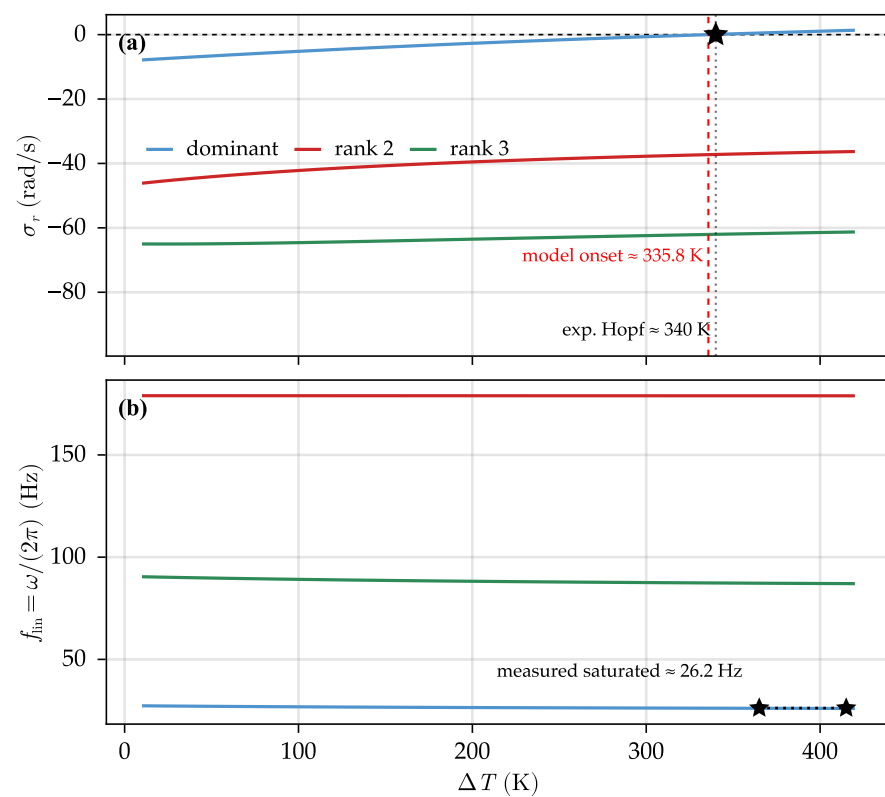
Envelope-residual spectrograms, all six conditions



Residual power below ≈ 0.1 Hz at every condition, strongest in the early growth phase.

BACKUP

Operator spectrum and mode placement

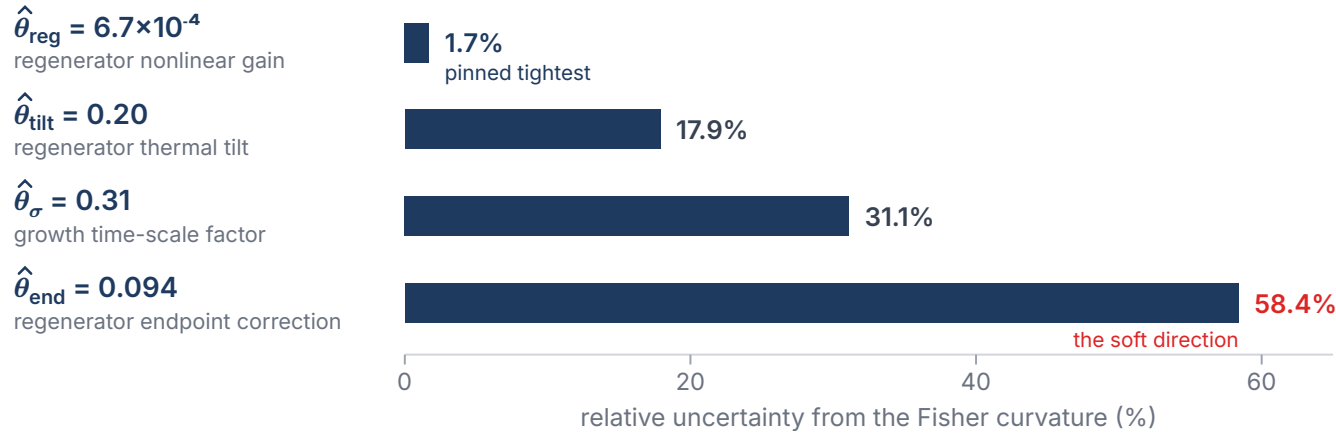


What the identification delivers

identified value

how tightly the data pin it

shorter = pinned harder



Identified, and error-barred, on the genuine forward.

- The numbers are the **identified values** $\hat{\theta}$, the promised coefficients read off the fit. θ_{curve} identifies at ≈ 0 , and the correlation map below shows it sitting nearly orthogonal to the rest.
- The bars are **how tightly the data pin each one**: the Fisher curvature of the fit, shorter is better. Mechanics in the backup.
- The loss level is pinned to **1.7%**. The endpoint correction is the soft direction at 58%.

θ_{reg}	+1.00	-0.80	+0.62	+0.90	-0.12
θ_{tilt}	-0.80	+1.00	-0.85	-0.92	+0.24
θ_{end}	+0.62	-0.85	+1.00	+0.72	-0.17
θ_{σ}	+0.90	-0.92	+0.72	+1.00	-0.22
θ_{curve}	-0.12	+0.24	-0.17	-0.22	+1.00
	θ_{reg}	θ_{tilt}	θ_{end}	θ_{σ}	θ_{curve}

Parameter correlations, genuine-forward covariance.

BACKUP

The loudspeaker sector, identified alongside

Coefficient	Role	Pinned by
θ_{det}	baseline frequency detuning	frequency residual
θ_{joint}	endpoint-sensitive coupling detuning	detuning residual
$\theta_{\text{hot,det}}$	hot-end detuning correction	frequency residual, hot cases
$\theta_{\text{hot,Z}}$	hot-end complex-load dissipation	saturated tail
θ_{hot}	boundary volume-velocity coupling	loudspeaker mobility u'/p'
κ_Z	load-to-detuning proportionality	held fixed for identifiability

Amplitude enters the speaker sector at envelope level: the hot-end load and detuning corrections scale with $x_{\text{pk}} \propto |A|$. The linear Z_{sp} row of $\mathbf{L}$ is not modified.

- Identified **together with the regenerator four**: the adjoint returns all ten gradients in one backward sweep.
- In the ten-parameter Fisher the **detuning components are the sharpest-determined directions** (top three eigenvalues). They pin the frequency and mobility channels.
- The **saturated pressure branch** is carried by the regenerator four, which is why the talk's axis stays there. u' remains the independent check channel.
- κ_Z is fixed, not fitted, and the genuine forward does not contain it at all.

The machine

"Working gas and pressure?"

air · 101.3 kPa

Ambient rig, cold end near 298 K. Low cost by design: the method is the point.

"Regenerator details?"

#60 · 20 mm · φ 0.73

Stacked screens, wire 0.15 mm. 100 nodes with the Ueda (2009) radius $r_0 = 0.123$ mm.

"Electric power? Efficiency?"

not optimised

The $500\ \Omega + 1\ \mu\text{F}$ load is a clean, characterised boundary for identification. Load design comes after, from the operator.

"Two sensors for ten parameters?"

2 pts × 6 ramps

The operator computes the mode shape, the sensors pin two points, the transients give thousands of samples. Fisher decides.

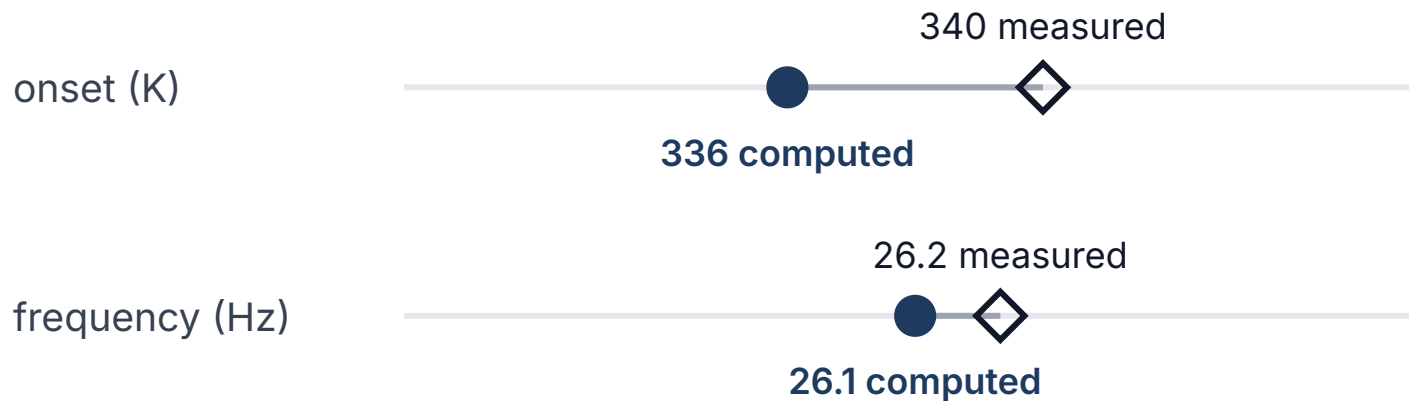
"Temperature profile inside the regenerator?"

ends only

Only measured end temperatures are imposed. The internal dynamics return as the missing state $\bar{T}_{\text{reg}}(t)$.

The linear operator

"How good is it, really, with zero knobs?"



Measured inputs only: geometry, gas, end temperatures, bench-measured Z_{sp} . Zero tuned knobs.

"Which Rott functions? Grid convergence?"

145 nodes

Boundary-layer forms in tubes, porous Ueda form in the screens. Hessian stable to 5 digits under a 10^4 step change.

"Why is 26 Hz the right mode?"

$$< -36 \text{ s}^{-1}$$

The runner-up modes never approach crossing over the whole heating range (operator-spectrum backup).

"Why the 336 vs 340 K gap?"

$$\approx 1\%$$

Thermocouple-to-screen heat leak and the imposed end profile. Conservative direction: the threshold is under-predicted.

The identification

"Is $\Sigma_{reg}(Re)$ another fitted curve?"

derived · $\propto Re_h$

Tanaka's oscillatory screen friction, $0.8\nu Re_h/d_h^2$, at the operator's own screen velocity. $\hat{\theta}_{reg}$ only sets its size, $\approx 10\%$ from first principles (6.7 vs $\approx 6 \times 10^{-4}$).

"How are envelopes extracted?"

26 Hz carrier

Complex demodulation of both channels around the tracked carrier. Temperatures prescribed as measured.

"Eight channels, fixed weights. Sensitive?"

path $\neq$ destination

Weights shape the warm start only. Every reported number closes on the branch metric, where weights do not enter.

"How is the measured amplitude defined?"

$\pm 10\%$, together

Other plausible definitions move model and measurement together. The $\approx 2\%$ closure survives.

"Only six conditions?"

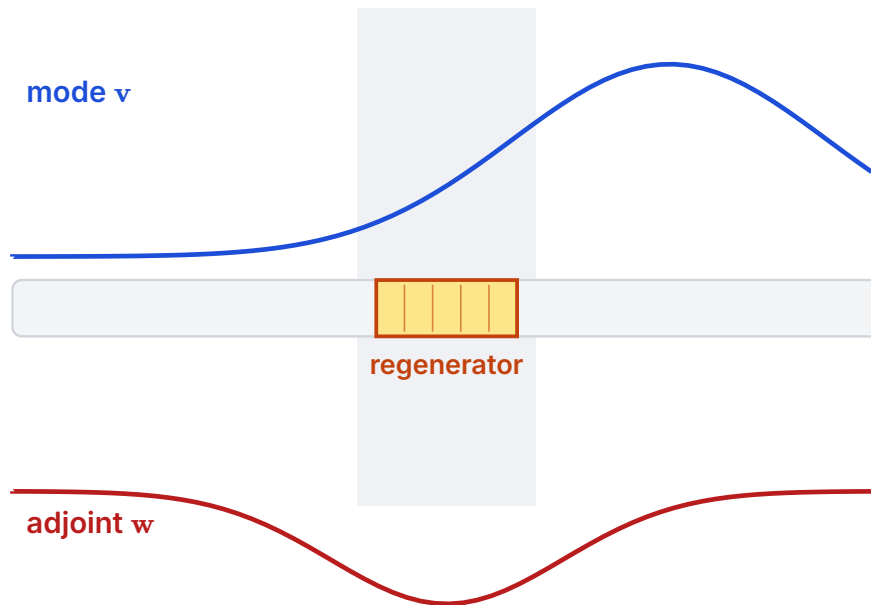
θ_{end} 58%

They span the branch and leave-one-out transfers. More data would mainly sharpen the softest direction.

Why the loss can be a product

$$\delta(i\sigma) = \mathbf{w}^H \delta \mathbf{L} \mathbf{v} \Rightarrow \sigma_{\text{reg}}^{\text{nl}} = \underbrace{\theta_{\text{reg}}}_{\text{magnitude}} \cdot \underbrace{g(\Delta T)}_{\text{branch trend}} \cdot \underbrace{\Sigma_{\text{reg}}(Re_h)}_{\text{amplitude law}}$$

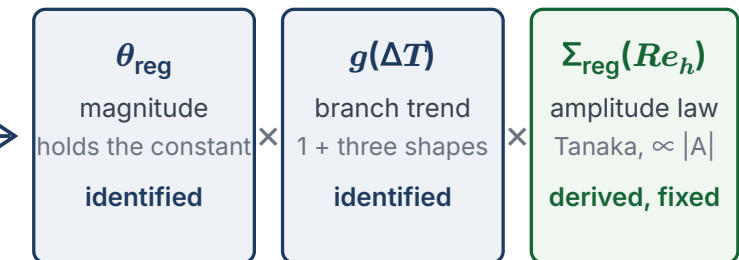
the friction always lives here, same spatial pattern



$\mathbf{w} \cdot \delta \mathbf{L} \cdot \mathbf{v}$ collapses
to ONE overlap
constant

only the strength changes,
with $|A|$ and ΔT

separable: $|A|$ and ΔT each live in one factor



first order, like every adjoint result in this talk

1

Physics fixes the shape

Tanaka friction in the screens, $\propto |A|$

2

Projection makes it one scalar

fixed pattern, so one overlap constant

3

The data judge the ansatz

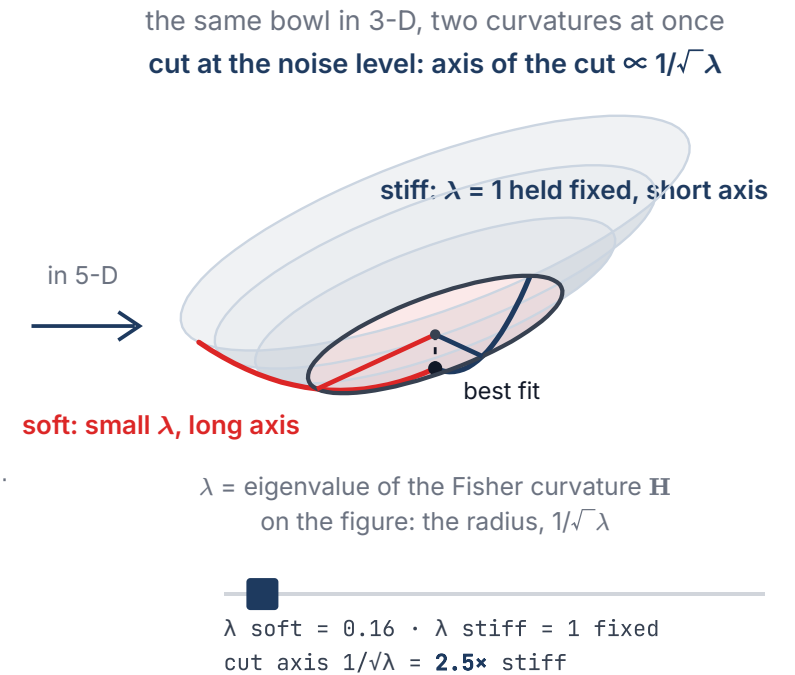
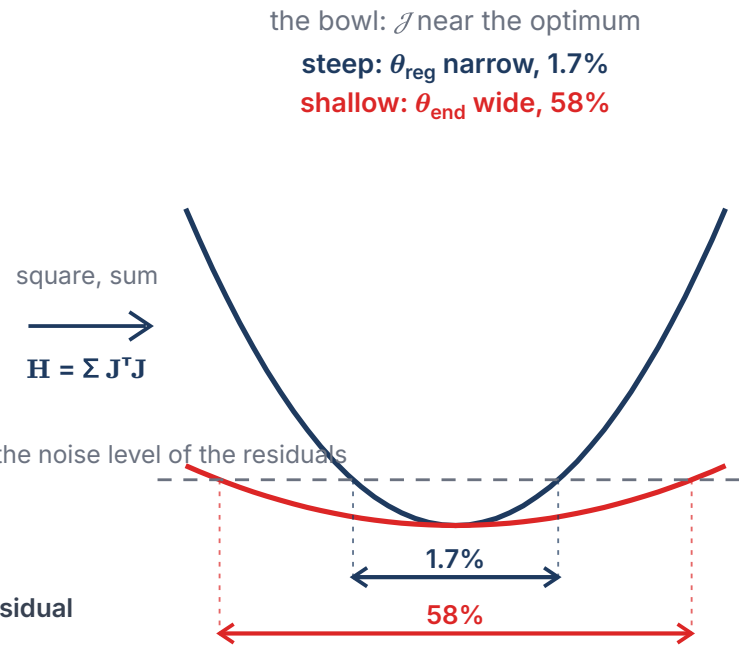
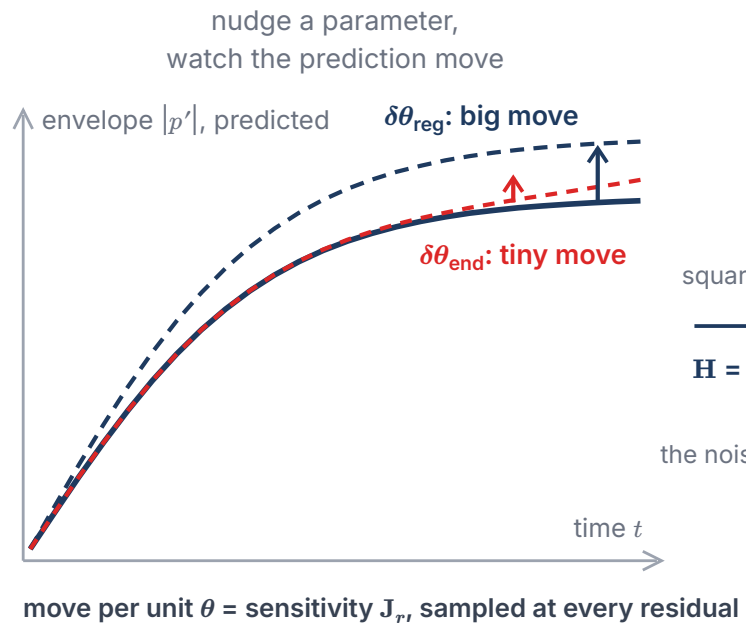
one θ set · full rank · LOO $\leq 2.4\%$

Where the error bars come from

$$\mathcal{J}(\hat{\theta} + \delta\theta) \approx \mathcal{J}(\hat{\theta}) + \frac{1}{2} \delta\theta^T \mathbf{H} \delta\theta, \quad \mathbf{H} = \sum_r \mathbf{J}_r^T \mathbf{J}_r \text{ (Gauss-Newton = Fisher),}$$

4), $\text{bar}_i = \sigma_i / \hat{\theta}_i$

$$\sigma_i = \sqrt{\hat{\sigma}^2 (\mathbf{H}^{-1})_{ii}}, \quad \hat{\sigma}^2 = \mathcal{J}(\hat{\theta}) / (N_c -$$



1

The fit lands in a bowl

gradient zero, curvature $\mathbf{H}$ from residual sensitivities

2

Width at the noise level

steep bowl = tight, shallow bowl = soft

3

Local diagnostic, stated as such

global checks: full rank + LOO

The two sweeps, concretely

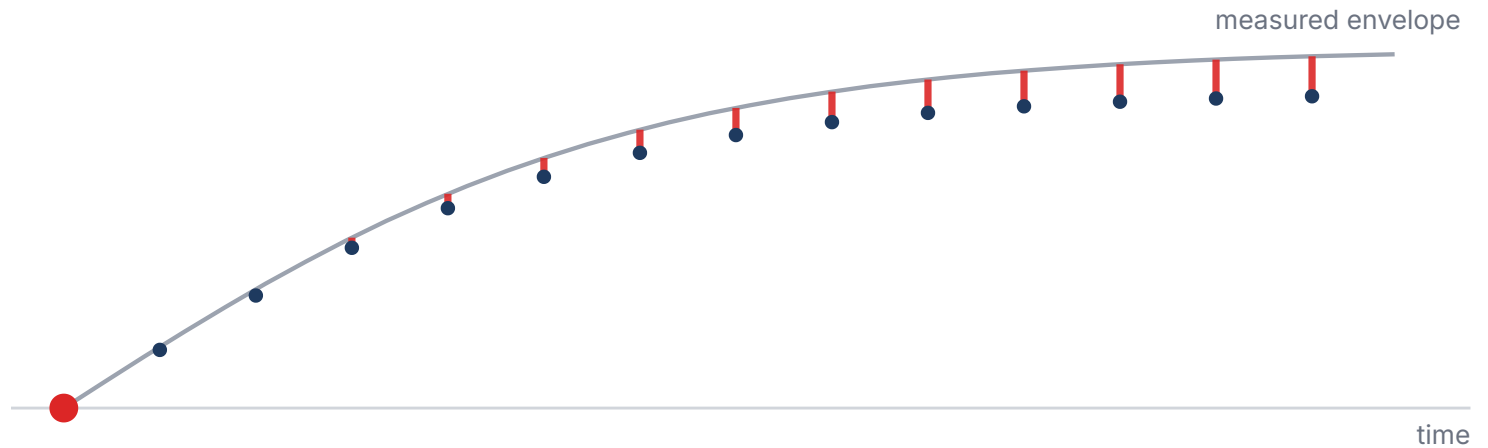
$A_{n+1} = f(A_n, \theta)$ run and store · $\lambda_n = (\partial f_n / \partial A_n)^\top \lambda_{n+1} + \partial \mathcal{J}_n / \partial A_n$ blame, carried backwards · $\nabla_{\theta} \mathcal{J} = \sum_n \lambda_{n+1}^\top \partial f_n / \partial \theta$ all parameters, one pass

$\nabla_{\theta} \mathcal{J}$, all five at once

θ_{reg}	
θ_{tilt}	
θ_{σ}	
θ_{end}	
θ_{spk}	

forward: run the model, store the path, measure the misfit

backward: carry the blame back, every parameter collects its share



1

Forward: run and store

the model marches, the misfit is measured

2

Backward: one pass of blame

λ carries $\partial \mathcal{J} / \partial \text{state}$ back through the stored path

3

Every parameter reads its share

cost independent of how many

Frozen vs genuine

"Why not an adjoint on the genuine forward?"

cheap + honest

The frozen gradient lands the warm start, a few dozen genuine runs reach the two-percent branch. A genuine-forward adjoint is natural future work.

"What does the genuine forward cost?"

80 s · every 20 steps

One run over all six conditions, operator re-solved every 20 steps, eigenvector tracked. Refinement in desktop minutes to hours.

"Eigenvalue sensitivities are not holomorphic."

d/d θ of the recursion

The gradient differentiates the real time-stepping map, never a complex derivative of $i\sigma$. The formula on the adjoint slide is intuition.

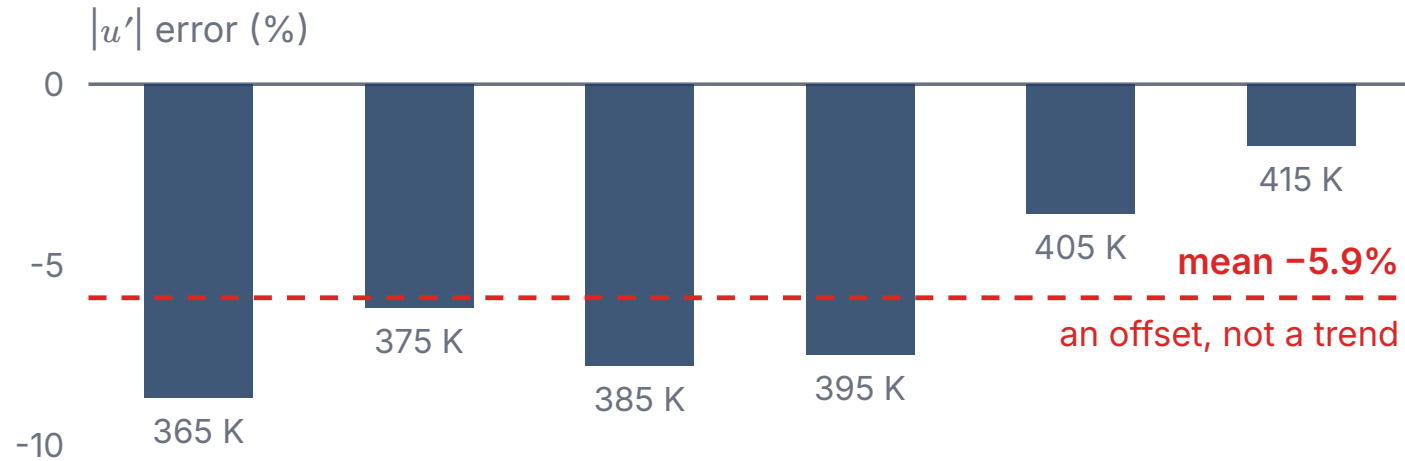
"Is κ_Z fitted?"

10 $\equiv$ 20 $\equiv$ 40

Fixed for identifiability. The genuine forward does not contain it: the published branch is bit-identical.

Validation

"Why is u' always about 6% low?"



A speaker-side underfit, not a shape error: the mobility gains stay at their warm-start values. Kept untouched as the held-out consistency check.

"Does leave-one-out survive the hottest case?"

0.08%

The 415 K fold is the best fold. Its refit sits on the full-data estimate.

"Supercritical? Any hysteresis?"

smooth at 340 K

Continuous branch, no jumps observed. The six transients are the quantitative constraint.

"Dropping θ_{curve} : tuning by the back door?"

$|\rho| \leq 0.24$

The data say we can. Fixing it to zero changes nothing else.

Scope and outlook

"Does this transfer to my engine?" The recipe:

1

Geometry + gas
for the operator



2

One boundary impedance
if a transducer is attached



3

Transients above onset
the data



4

Identified operator
loss basis re-chosen per
regenerator

"Could DeltaEC have done this?"

steady ≠ transient

DeltaEC answers the operating point. Here: transient data, adjoint gradient, identifiability. Complementary, and it is the quasi-linear baseline of the opening figure.

"This looks like 4D-Var."

deliberately

Linearised inner solve, nonlinear relinearisation. New here: brought to thermoacoustics with the surrogate-versus-genuine check.

"Did you dissociate the mechanisms, as Penelet prescribed?"

"A key point : each of the NL processes operates with its own time-scale."

His key point (Sendai 2014, slide 25), our measurement: the one mechanism with its own slow clock, the regenerator temperature field, separates below 0.1 Hz. The fast ones merge into the aggregate loss.

"Next steps?"

$\bar{T}_{\text{reg}}$ · adjoint · load

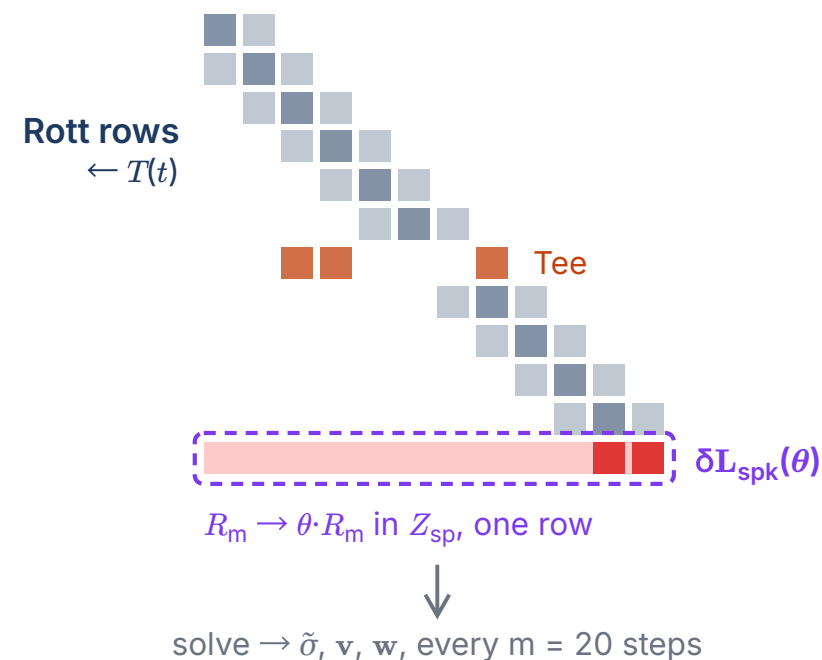
The temperature state, a genuine-forward adjoint, and design from the identified operator.

How $\delta \mathbf{L}(\theta)$ enters the matrix

$$(\mathbf{L}(T) + \delta \mathbf{L}) \tilde{\mathbf{x}} = i\tilde{\sigma} \mathbf{S} \tilde{\mathbf{x}}, \quad \text{re-solved at every refresh node}$$

- **Rott blocks, rebuilt at the measured temperatures.** Every refresh re-evaluates the tridiagonal rows M_j, K_j, G_j at $T_{\text{hot}}(t_k), T_{\text{cold}}(t_k)$. Temperature is what changes within a run.
- **Speaker row, the θ entries in the matrix.** $P_N = Z_{\text{sp}} U_N$ with $R_m \rightarrow \theta_{R_m} R_m$, and the rear-stiffness scale when the housing is modelled. This is $\delta \mathbf{L}_{\text{spk}}(\theta)$.
- **The loss shape stays outside the matrix.** After the re-solve, the branch-decisive corrections act on the projected growth rate, $\sigma_{\text{eff}} = \tilde{\sigma}_r - \theta_{\text{reg}} g(\Delta T) \Sigma_{\text{reg}}(Re_h) - \sigma_{\text{spk}}^{\text{nl}}$, with Re_h read from the *fresh* eigenvector's velocity ratio.

The frozen surrogate keeps $\sigma, \mathbf{v}, \mathbf{w}$ from one solve and applies $\mathbf{w}^H \delta \mathbf{L} \mathbf{v}$ with them. The genuine forward re-solves and re-projects, no expansion.



One refresh node: what is rewritten before the eigensolve.